

From Stage to System: Bias Propagation in Clinical AI Pipelines

Luwam Major Kefali
luwammajor.kefali@studio.unibo.it
0001130689

Hilina Fisssha Woreta
hilina.woreta@studio.unibo.it
0001143250

August 2026

Abstract

Most fairness research treats a clinical AI system as a single model, but real hospital pipelines operate as chains of decisions: a note gets read, a risk score gets computed, and a patient gets enrolled in a care program or excluded. This project builds a three-stage pipeline on MIMIC-IV (an LLM-based note extraction stage, an XGBoost readmission classifier, and a top-10% care management enrollment decision) to investigate whether fixing bias at one stage resolves it end-to-end or simply displaces it. We find that bias enters before the tabular stage even runs. Perturbing a single demographic token in a clinical note leaves the LLM’s extracted risk scores unstable by as much as 8 points on a 10-point scale. This structurally fragile failure undermines the core assumption of counterfactual fairness testing and cascades directly into downstream enrollment decisions. At the tabular stage, we test four established mitigation methods and demonstrate that one of them, counterfactual data augmentation, reports a perfect fairness score that is an artifact of testing itself on its own assumptions rather than a real fix. We then introduce two new methods that split model features into “real health” and “unequal access” groups, utilizing resampling from real patients and a formal causal model fit with DoWhy. Both approaches reduce individual-level unfairness by roughly 80% with under 1 point of AUROC lost. The resampling version successfully resolves the insurance-axis disparity and holds up on an independent real-patient validation check, though a separate latent-space evaluation reveals it as the weaker of the two, representing a genuine trade-off rather than a definitive solution. Finally, a jointly-trained pipeline-aware method produces a measurable but practically negligible and mildly harmful effect. This outcome serves as direct evidence for the bias migration hypothesis this project set out to test.

Contents

List of Acronyms	4
1 Introduction	5
2 Related Work	5
3 Data & Pipeline	6
3.1 Cohort and Feature Engineering	7
3.2 Stage 1: LLM-Based Clinical Note Extraction	7
3.2.1 Context Window Optimization	8
3.2.2 Mitigation of Zero-Shot Failure Taxonomies	8
3.2.3 Extraction Reliability and Early Fairness Audit	8
3.2.4 Distribution of Extracted Clinical Features	8
3.2.5 Internal Validation of Extracted Signals	9
3.2.6 Human-in-the-Loop Validation and Inter-Rater Reliability	11
3.3 Stage 2: Tabular Readmission Classifier	12
3.4 Calibration Quality and Interpretability	12
3.5 Stage 3: Care Management Enrollment	13
4 Fairness Methods	14
4.1 Counterfactual Fairness and CFVR	14
4.2 Counterfactual Generation and Clinical Semantic Preservation	14
4.2.1 The Perturbation Instability Phenomenon in LLMs	14
4.2.2 Template-Based Perturbation of Demographic Markers	15
4.2.3 The Semantic Consistency Problem and SBERT Validation	15
4.3 Bias Migration Ratio	16
4.4 Four Baseline Mitigation Methods	16
4.5 Pipeline-Aware Counterfactual Fairness Training	17
4.6 Path-Aware Counterfactual Data Augmentation	18
4.7 DoWhy Structural Causal Model Counterfactuals	18
5 Results	19
5.1 Main Cohort Comparison	19
5.2 Statistical Significance and the Insurance Axis	19
5.3 MPPD: Real-Patient Validation	21
5.4 Stage 1 Sub-Analysis: Adding LLM Features	21
5.5 Level 2: Latent-Space Counterfactual Fairness	23
5.6 Intersectional Fairness: Race x Sex and Race x Insurance	23
5.7 Bias Trace Visualizer and Conformity Assessment	25
6 Ethical Analysis	27
7 Limitations and Future Work	28
8 Conclusion	29
A LLM Prompt Template and Pydantic Schema	32
A.1 Pydantic Validation Schema	32
A.2 System Prompt Template	32

B	Human Annotation Guidelines	33
B.1	Annotation Rubrics	33
B.2	Inter-Rater Reliability Formulation	33
C	Counterfactual Perturbation Examples	34
D	XGBoost Hyperparameter Search Space	34
E	Complete Tabular Feature Dictionary	34
F	Baseline Cohort Characteristics	35

List of Acronyms

Acronym	Meaning
AUROC	Area Under the Receiver Operating Characteristic curve
BMR	Bias Migration Ratio
BUN	Blood Urea Nitrogen
CDA	Counterfactual Data Augmentation
CFVR	Counterfactual Fairness Violation Rate
dp_ratio	Demographic Parity Ratio
ECE	Expected Calibration Error
ICD	International Classification of Diseases (diagnosis coding system)
LLM	Large Language Model
MIMIC-IV	Medical Information Mart for Intensive Care IV
MPPD	Mean Predicted Probability Difference
p_{adj}	Benjamini–Hochberg adjusted p-value
SBERT	Sentence-BERT
SCM	Structural Causal Model
SDOH	Social Determinants of Health
SDPA	Scaled Dot-Product Attention
SHAP	SHapley Additive exPlanations
VAE	Variational Autoencoder

1 Introduction

In 2019, Obermeyer et al. showed that a widely used care management algorithm underestimated illness severity in Black patients. The failure wasn't in any single broken component: the whole system used healthcare cost as a proxy for health need, and cost tracks access more than it tracks sickness [12]. That's a pipeline problem, not a model problem, and it's also why healthcare is the flagship high-risk category under the EU AI Act: conformity assessments there will need to evaluate systems, not just models in isolation.

That distinction, system versus model, is the gap most single-stage fairness interventions leave open, and it's the one this project is built to test. The central question: when a standard fairness fix is applied to one stage of a clinical pipeline, does it actually reduce discrimination by the time a patient reaches the final decision, or does the bias just resurface somewhere downstream? We call this the bias migration hypothesis. To test it, we built a three-stage pipeline on MIMIC-IV, deliberately mirroring the Obermeyer et al. setup: an open-weight LLM reads discharge notes and extracts structured risk signals, an XGBoost model turns tabular features (plus those extracted signals) into a calibrated readmission risk score, and a threshold rule enrolls the top 10% of patients.

Against that setup, this paper makes four contributions. First, we show that the Stage 1 LLM extraction stage itself is unstable under demographic perturbation: swapping a single demographic token in a clinical note leaves the model's extracted risk scores fluctuating by as much as 8 points on a 10-point scale, a symmetric but structurally fragile failure that complicates the very idea of testing counterfactual fairness on generated text. Second, we empirically test four established mitigation methods, plus a jointly-trained pipeline-aware method, against the bias migration question, using a formal Bias Migration Ratio (BMR) metric. Third, after the jointly-trained method turned out to have a real but negligible effect, we construct two new mitigation methods that separate a patient's features into what is plausibly real biology and what is plausibly the result of unequal access, and intervene only on the latter. Fourth, we show that one of the established methods, counterfactual data augmentation, can report a perfect individual-fairness score that is actually an artifact of evaluating it against the exact assumption it was built on, and that an independent check catches this where the standard metric does not.

2 Related Work

Kusner et al. define counterfactual fairness as the requirement that a model's prediction for an individual shouldn't change if that individual's protected attribute had been different, holding everything not caused by the protected attribute fixed [8]. We call this the Level 1 definition, and it's the one our main CFVR metric is built on directly: swap the protected attribute, hold every other observed feature fixed, and check whether the decision changes. The practical problem with the naive version of this idea is that it treats every feature correlated with the protected attribute as something to erase, including features that reflect real biological differences that have nothing to do with discrimination.

This project also builds a second, independent version of the same test, Level 2, that swaps the protected attribute inside a learned latent representation rather than the raw observed features (Section 5.5). Level 2 was specified in the original project proposal alongside Level 1, but its purpose sharpened only once Level 1 results were in hand: Level 1's own construction bakes in an assumption, that holding every other observed feature fixed is the right counterfactual, and CDA's Level 1 score in particular turns out to depend on that assumption in a way that made an independent, differently-constructed check necessary (Section 5.1). Level 2 is that check.

Path-specific fairness work addresses exactly that problem. Nabi and Shpitser argue that not every causal pathway from a protected attribute to an outcome is illegitimate, some pathways are

the ones you actually want to preserve, and fairness interventions should target the illegitimate pathways specifically rather than nulling the whole effect [11]. Chiappa extends this into path-specific counterfactual fairness with a more general causal framework [3], and Nabi and Benkeser frame fair risk minimization directly as a constrained optimization over path-specific effects [10]. Kilbertus et al. introduce the idea of a resolving variable, a feature that is influenced by the protected attribute in a way that’s accepted as legitimate rather than discriminatory, using university admissions and department choice as the classic example [7]. This is an explicit ethical judgment call baked into the model, not something the data itself can settle, and it’s the exact idea our path-aware methods are built on.

There is real-world precedent for revisiting which uses of a protected attribute count as legitimate. Vyas, Eisenstein and Jones document how race correction formulas in medicine, in kidney function estimates and in VBAC risk calculators, were built on the assumption that race reflects biology, and how that assumption has since been rejected because race in those formulas was really tracking structural inequality [14]. That’s the same distinction we’re trying to draw between real health features and access-related features, just playing out in actual clinical practice rather than in a fairness paper.

On the mitigation side we use four standard baselines. Kamiran and Calders reweight training examples so the joint distribution of label and protected attribute is balanced [6]. Hardt, Price and Srebro’s equalized odds post-processing adjusts decision thresholds per group after training, without touching the model itself [4]. Zhang, Lemoine and Mitchell’s adversarial debiasing trains a predictor alongside an adversary that tries to recover the protected attribute from the predictor’s internal representation [17]. The fourth is counterfactual data augmentation (CDA), which builds a race-swapped twin of every training patient and adds it to the training set, Section 4.4 covers it in more detail since our two novel methods are both direct responses to a specific problem with it. Later we also use a real-patient auditing approach adapted from Xu, Hwang et al., who match individuals across demographic groups by clinical similarity rather than by synthetic counterfactuals, which is what our MPPD cross-check is built on [16].

3 Data & Pipeline

We build a three-stage clinical pipeline on MIMIC-IV [5], a publicly available, de-identified database of intensive-care and hospital admissions: a note gets read, a risk score gets computed, and an enrollment decision gets made, the same shape as the care-management system Obermeyer et al. studied [12]. Most of the tabular data this project runs on already exists in MIMIC-IV’s own structured tables (labs, ICD diagnosis codes, demographics), independent of anything the note-reading stage does. Stage 1 only adds nine extra features pulled out of unstructured discharge notes on top of that already-structured base. It does not create the tabular data from scratch.

Three separate models get trained over the course of this project, reflecting a practical constraint on Stage 1’s scale. Running the Stage 1 LLM extraction pipeline across the full cohort of roughly 280,000 admissions was not computationally feasible within this project’s scope, so LLM extraction was carried out on a smaller sub-cohort of 3,995 patients instead. The main model, the Stage 2 XGBoost classifier described in Section 3.3, is trained on the full cohort using all thirty-five tabular features available for every admission; every fairness method in Section 4 and every result in Section 5 is tested against this one model, and it is by far the most heavily used model in this report. Section 5.4 introduces two additional, much smaller models, called Model A and Model B there, trained from scratch on the 3,995-patient sub-cohort where Stage 1 LLM extractions are available: Model A uses the same thirty-five tabular features as the main model, Model B adds the nine LLM-derived features on top. Holding the patient population fixed between A and B isolates the effect of adding LLM-derived features specifically, rather than conflating it with a difference in cohort. The same core fairness metrics

used elsewhere in this report, demographic parity and per-group enrollment, TPR, and FPR, are computed for both models, with full results reported in Section 5.4.

3.1 Cohort and Feature Engineering

Everything downstream of Stage 1 runs on 35 tabular features pulled from MIMIC-IV: age, sex, insurance type, admission type, length of stay, nine lab values (bicarbonate, BUN, creatinine, glucose, hemoglobin, platelets, potassium, sodium, and white blood cell count), sixteen comorbidity flags, a comorbidity count, two ICU utilization features (total ICU length of stay and number of ICU stays), and the number of prior admissions in the twelve months before the index visit. The sixteen comorbidity flags are not free-text fields. They are built by mapping each admission’s ICD-9 and ICD-10 diagnosis codes onto sixteen clinical categories (congestive heart failure, arrhythmia, hypertension, chronic pulmonary disease, diabetes, renal failure, liver disease, cancer, obesity, depression, anxiety, alcohol abuse, drug abuse, psychosis, coagulopathy, and HIV/AIDS), so a flag is 1 only if a matching diagnosis code actually appears on that admission. Prior admissions is computed with a self-join against the same table: for every admission, count how many other admissions by the same patient happened in the preceding 365 days. The target label is thirty-day all-cause readmission.

Lab values are handled on a complete-case basis rather than imputed: an admission missing any of the nine lab values is dropped from the cohort instead of having the gap filled in. The reasoning follows the same logic used later in the fairness analysis: a filled-in value is a guess, and guesses can systematically favor or disadvantage a group depending on who tends to be missing which labs. So the safer choice is to only model patients for whom the real values exist, while checking that dropping them isn’t concentrated in one racial or insurance group.

Race is normalized into five categories (White, Black/African American, Hispanic/Latino, Asian, Other/Unknown) and insurance into a fixed set of categories, both stored as integer codes fixed once in a saved encoding file and reused everywhere downstream rather than re-derived on the fly. Fixing the codes this way rules out a specific failure mode: an encoder that assigns codes based on whatever categories happen to appear in a given run can silently change what a given code means from one run to the next. Two intersectional columns, race combined with sex and race combined with insurance, are also constructed at this stage for the intersectional fairness checks in Section 5.

The train, validation and test splits are done at the patient level with scikit-learn’s `GroupShuffleSplit` on `subject_id` (70/15/15), not at the admission level. A patient with two admissions could otherwise land in both the training set and the test set, which would let the model partly memorize that specific patient instead of learning something that generalizes, and would make test performance look better than it really is. This is checked explicitly with a patient-overlap assertion on every split, confirming the three sets share no patients. The resulting splits are 279,569 training admissions, 60,391 validation admissions, and 59,409 test admissions.

3.2 Stage 1: LLM-Based Clinical Note Extraction

The initial stage transforms unstructured MIMIC-IV discharge summaries into structured features for downstream classification. We deployed an open-weight large language model, Llama-3.1-8B-Instruct, configured with 4-bit quantization and scaled dot-product attention (SDPA) to produce structured JSON outputs. The extraction targeted four specific prognostic domains: comorbidity burden scores, social determinants of health (SDOH) flags, psychiatric complexity, and discharge risk indicators.

3.2.1 Context Window Optimization

Clinical discharge summaries in MIMIC-IV frequently exhibit significant length variance. To balance the preservation of clinical context with the computational constraints of large-scale batch inference, a targeted truncation strategy was implemented. Notes were restricted to a maximum of 6,000 characters (capturing the first 3,000 and final 3,000 characters). This specific heuristic successfully preserved critical admission details and the “Brief Hospital Course” sections while maintaining a predictable prefill token budget.

3.2.2 Mitigation of Zero-Shot Failure Taxonomies

Early iterations of the zero-shot extraction pipeline revealed distinct failure modes that could introduce systemic, non-demographic artifacts into the downstream tabular classifier. The model exhibited categorical mode collapse in psychiatric complexity classification and “cart before the horse” hallucinations, where SDOH flags were inferred from implied demographics rather than explicit textual evidence.

To resolve these failures prior to counterfactual fairness testing, structural constraints were enforced. The Pydantic extraction schema was reordered to mandate the extraction of textual evidence prior to boolean SDOH classification, anchoring the model’s logic to the clinical text. Furthermore, a finite-state machine constraint was applied during autoregressive generation using the Outlines library. This approach strictly enforced the target schema at the token level, guaranteeing 100% valid JSON syntax and completely eliminating categorical spillover across the full inference batch. The outputs of this optimized extraction process were subsequently validated against manual annotations on a held-out subset.

3.2.3 Extraction Reliability and Early Fairness Audit

In the context of the bias migration hypothesis, algorithmic disparities can manifest as early as the natural language processing stage if an extraction model performs inconsistently across protected demographic groups. To validate the integrity of the Stage 1 pipeline, overall extraction reliability and subgroup fairness were audited.

The constrained generation approach utilizing the `outlines` library yielded a 99.9% overall success rate, with only 5 records failing validation out of the 4,000 sampled admissions. Crucially, as detailed in Table 1, there was no significant variance in extraction success or plausibility across racial demographics. This confirms that the LLM extraction layer did not introduce early pipeline bias via systemic failure rates for minority groups.

Table 1: Stage 1 LLM Extraction Reliability and Plausibility by Race

Race / Ethnicity	Sample Size (n)	Success Rate (%)	Plausibility Rate (%)
Asian	124	100.0	100.0
Black/African American	579	100.0	100.0
Hispanic/Latino	198	100.0	100.0
Other/Unknown	297	100.0	100.0
White	2802	99.8	99.8
Total Cohort	4000	99.9	100.0

3.2.4 Distribution of Extracted Clinical Features

A primary failure mode of zero-shot large language models in constrained extraction tasks is mode collapse, where the model defaults to a single, safe, or average response across all inputs.

To verify generation variance, the distributions of the extracted variables were analyzed across the cohort.

The resulting distributions demonstrate appropriate scaling of clinical risk. The comorbidity burden scores and discharge risk indicator counts form healthy distributions, indicating the model successfully differentiated between low, medium, and high-complexity patients based on the provided text. Similarly, the incidence rates for the social determinants of health (SDOH) capture realistic socio-economic variations without evidence of systemic over-flagging.

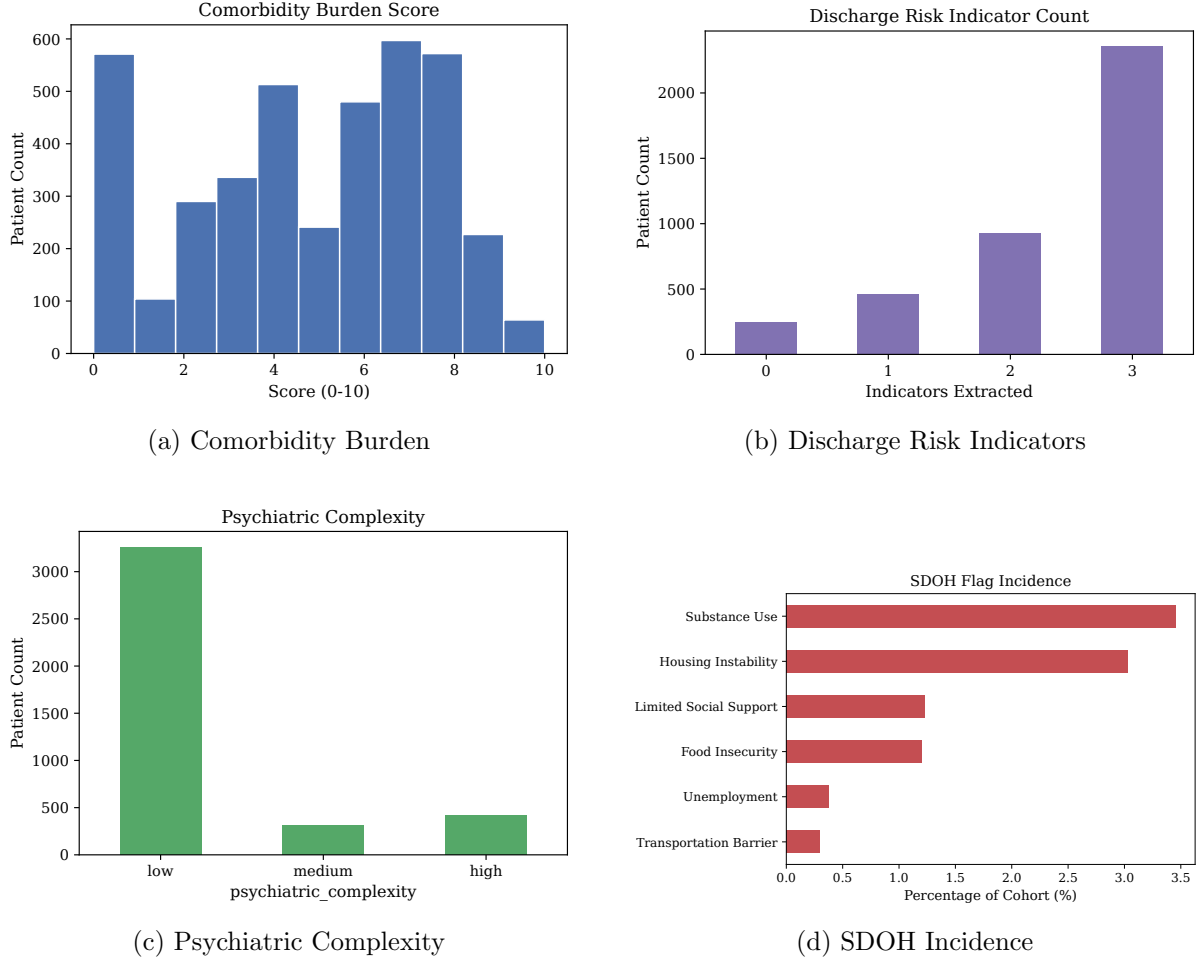


Figure 1: Distributions of structured outputs extracted by the LLM from the unstructured MIMIC-IV discharge summaries. The presence of broad variance confirms the model successfully avoided mode collapse.

3.2.5 Internal Validation of Extracted Signals

Before passing the extracted features to the downstream tabular classifier, an internal validation was conducted to ensure the LLM outputs adhered to expected clinical and statistical realities. This acts as a secondary check against hallucination and mode collapse.

First, a Spearman rank correlation matrix was computed to evaluate feature independence. The resulting matrix confirms the LLM successfully disentangled medical complexity from socioeconomic risk. The extracted comorbidity burden score exhibits near-zero correlation (ranging from 0.01 to 0.06) with all social determinants of health (SDOH) flags. Conversely, the SDOH variables cluster logically: limited social support shows a strong positive correlation with both unemployment (0.44) and transportation barriers (0.41). Psychiatric complexity

also demonstrates a positive correlation with housing instability (0.19), accurately reflecting documented vulnerabilities in clinical populations.

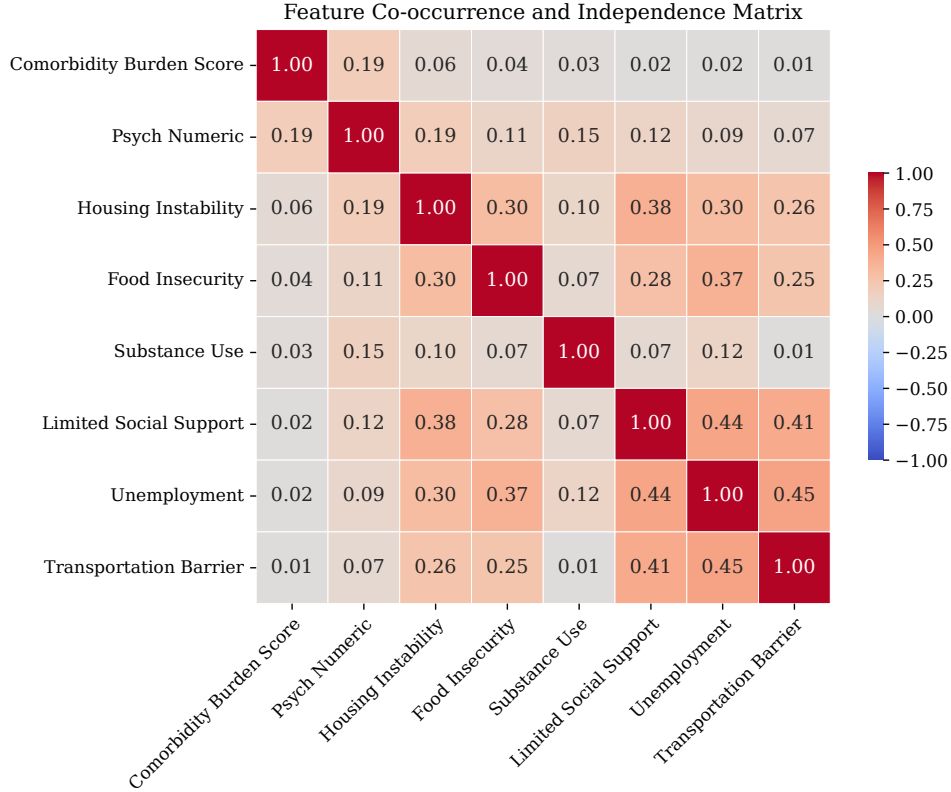


Figure 2: Spearman correlation matrix of extracted LLM features. Medical complexity (Comorbidity Burden Score) remains statistically independent from socioeconomic access barriers, validating the prompt’s isolation instructions.

Second, n-gram frequency analysis on the raw extracted text fields confirms the model grounded its output strictly in medical pathology. The most frequently extracted bi-grams and tri-grams include “heart failure,” “atrial fibrillation,” and “acute kidney injury,” verifying the absence of non-clinical filler text.

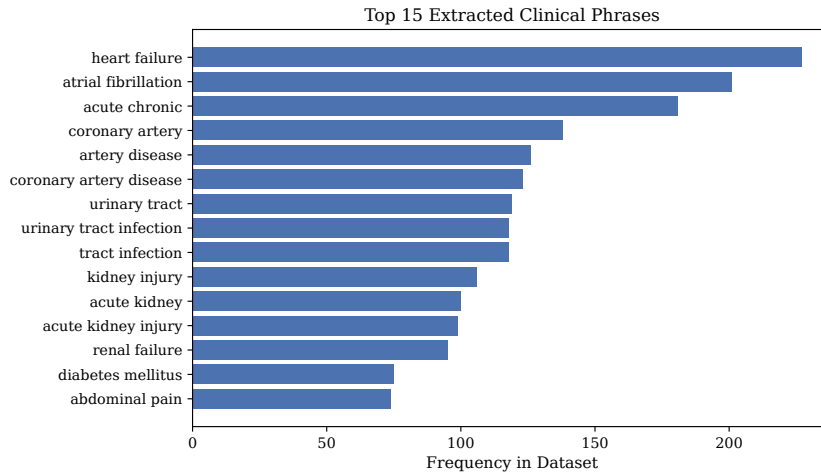


Figure 3: Frequency of the top 15 extracted clinical bigrams and trigrams. The terminology indicates strong adherence to clinical reality without hallucinated conversational filler.

Finally, a hexbin density analysis of note length versus extraction count confirms that the 6,000-character context window limit did not induce truncation bias. Clinical notes exceeding the length threshold consistently yielded standard extraction volumes without artificial capping.

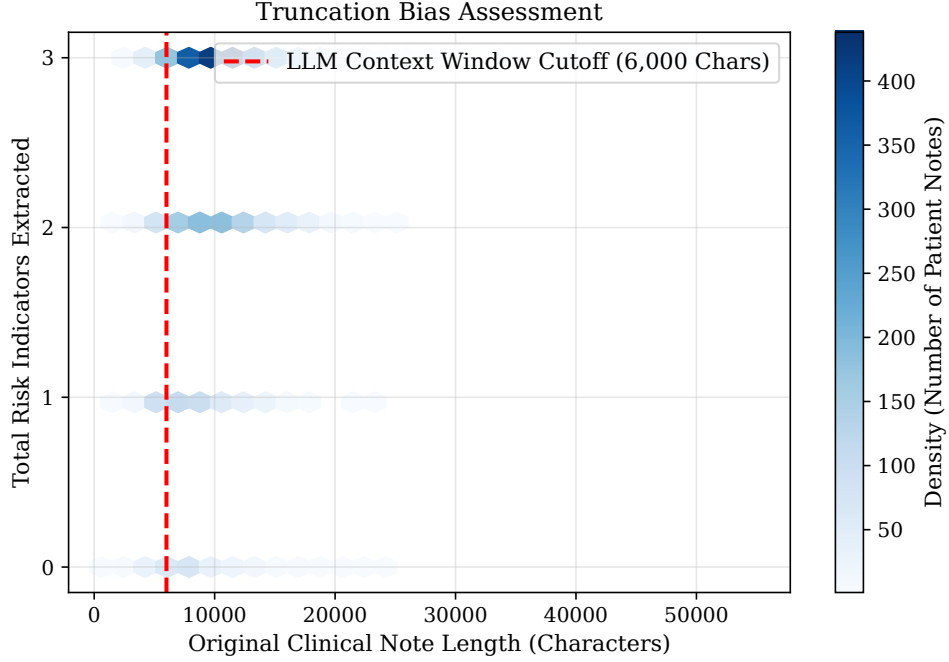


Figure 4: Hexbin density plot assessing truncation bias. Extraction volumes remain consistent even for clinical notes extending beyond the strict 6,000-character truncation threshold.

3.2.6 Human-in-the-Loop Validation and Inter-Rater Reliability

To confirm algorithmic extraction reliability, a stratified sample of 41 clinical discharge summaries underwent manual chart review by a domain expert. The human annotator utilized the exact grading rubric provided to the LLM in the system prompt.

To account for the varying data types produced by the schema, inter-rater alignment was evaluated across multiple prognostic domains. For the ordinal Comorbidity Burden Score, agreement was measured using Cohen’s Quadratic Weighted Kappa to penalize the magnitude of ordinal disagreements. The evaluation yielded a Kappa of 0.766, indicating substantial agreement between the machine extractions and human chart review.

For binary categorical Social Determinants of Health (SDOH) flags, standard precision and recall metrics become mathematically undefined when the true positive count in a small validation sample is near zero. Consequently, we evaluated rare flags (such as substance use) using the Exact Match Rate and Specificity (True Negative Rate). The extraction pipeline demonstrated a 92.7% exact match rate and a 97.4% specificity against human annotations, confirming the model’s high resistance to false-positive hallucinations when explicit textual evidence is absent.

Furthermore, subjective classification of Psychiatric Complexity achieved a 75.6% exact match rate across low, medium, and high categories, while the extraction volume for primary Discharge Risk Indicators tracked human counts with a Mean Absolute Error (MAE) of 1.27. Table 2 provides a representative sample of the factual ground-truth annotations compared against the automated LLM extractions, demonstrating high structural and clinical alignment.

Table 2: Sample comparison of Human vs. LLM structured extractions.

Admission ID	Human Score	LLM Score	Human SDOH	LLM SDOH
20131009	1	0	no evidence	no evidence
25216450	6	7	substance_use, lack-of-support	no evidence
24347789	1	2	no evidence	no evidence

3.3 Stage 2: Tabular Readmission Classifier

Stage 2 is an XGBoost classifier [2], a gradient-boosted decision tree model, trained on the thirty-five tabular features to predict thirty-day readmission. About 21% of admissions in the training set are actually readmitted within thirty days, so `scale_pos_weight` is set to the ratio of negative to positive examples during training; without it, a model chasing overall accuracy could just predict “not readmitted” for everyone and still be right most of the time. Hyperparameters (number of trees, learning rate, tree depth, subsampling rates) are tuned with Optuna [1] over 100 trials, scored by validation AUROC. A first pass with reasonable default parameters reached 0.6995 validation AUROC; tuning improved that to 0.7006. We did not test the 0.0011 gap for significance and don’t treat it as a meaningful improvement over the defaults.

Raw XGBoost outputs are not well-calibrated probabilities; before calibration the model’s mean predicted probability on the validation set was 0.455, more than double the actual 21.1% observed readmission rate. That gap matters because Stage 3 turns these scores into an enrollment decision, so a score of 0.3 should mean something close to a real 30% chance of readmission, not just a relative ranking of who is riskier than whom. Isotonic regression, fit on the validation set, closes this gap by mapping raw scores to calibrated probabilities: after calibration the mean predicted probability matches the observed rate exactly (0.211 vs. 0.211). Calibration uses isotonic regression rather than Platt scaling, a logistic-regression-based alternative that assumes the miscalibration follows a particular sigmoid shape. Isotonic regression makes no such assumption; it just fits a non-decreasing step function directly to the data, which matters here since there was no reason to expect this model’s miscalibration to happen to follow a sigmoid.

3.4 Calibration Quality and Interpretability

Calibration was checked per demographic group, not just overall, since a model can look well-calibrated on average while being systematically overconfident or underconfident for one specific group. For this check we set a per-group calibration target of $ECE < 0.05$; anything under that we treat as adequately calibrated for that group. Table 3 shows Expected Calibration Error (ECE) by race on the baseline model. Every group falls well under that 0.05 target, and White patients, the largest group by far, have the lowest ECE of any group (0.0023).

Table 3: Baseline model ECE by race

Group	n	ECE
White	41320	0.0023
Black/African American	9102	0.0118
Asian	2008	0.0129
Other/Unknown	4864	0.0131
Hispanic/Latino	3097	0.0160

SHAP (SHapley Additive exPlanations) values were computed on the baseline model for interpretability [9]. SHAP assigns each feature a signed contribution toward or away from a specific patient’s prediction, borrowed from a concept in cooperative game theory about how to

fairly split a payout among players who contributed unequally, which is what makes it useful here: it doesn't just say a feature mattered on average, it says how much it moved this particular patient's score, consistent with EU AI Act Article 13 documentation requirements. `race_enc` ranks 8th of 35 features by mean absolute SHAP value (0.0515), ahead of platelets, white blood cell count, and several other lab values, meaning it is not a negligible input to the model even before any mitigation is applied. This is a model-internal confirmation, independent of CFVR or `dp_ratio`, that race is doing real work inside the baseline model, not just correlating with the outcome incidentally.

The direction of that effect is not the same for every group. Mean `race_enc` SHAP is positive for White patients (+0.0320, pushed toward enrollment) and negative for every other group: Asian (−0.0075), Black/African American (−0.0146), Hispanic/Latino (−0.0223), and Other/Unknown (−0.2800, far larger in magnitude than the rest). Taken at face value this looks backwards, Black and Hispanic patients are actually enrolled at higher rates than White patients (Section 3.5), yet the race feature itself is pushing their scores down, not up. The resolution is that SHAP for one feature measures its marginal contribution after every other correlated feature has already had its say, and it is not the same thing as the raw group-level outcome gap. Checking the model's top features by race clarifies part of this: Black patients average 1.764 prior admissions in the past 12 months versus White patients' 1.217, and a comorbidity count of 2.883 versus 2.535, both higher, and `prior_admissions_12m` alone is the single largest driver of the model's predictions by a wide margin. Once that feature and comorbidity count have already pushed a Black patient's score up, there is less left for `race_enc` itself to explain, and what is left over points slightly the other way, this is not evidence the model is penalizing Black patients overall, since their higher real-world enrollment rate is being carried by genuine clinical risk factors rather than by the race feature amplifying it. The same check is weaker for Hispanic patients: their prior admissions are higher than White's (1.505) but comorbidity count, ICU stays, and length of stay are not, so this explanation only partly applies there. It does not apply cleanly at all to two of the five features checked, Black and Hispanic patients both have fewer ICU stays and shorter hospital stays than White patients, the opposite of what a straightforward severity story would predict, so this account is real but partial, not a complete explanation.

Other/Unknown's outlier SHAP value has a cleaner story. That group averages only 0.656 prior admissions, less than half of every other group's average and by far the lowest, which lines up directly with both its large negative SHAP and its low baseline enrollment rate (Section 3.5). It does, however, have noticeably more ICU stays (0.458 vs. 0.17–0.23 elsewhere) and a longer average length of stay (7.425 days vs. roughly 5.3–5.9 elsewhere), so it remains a clinically distinct group, and MIMIC-IV's "Other/Unknown" race label is itself a catch-all category covering declined-to-state, genuinely unknown, and multi-racial patients, which may be doing some of that work rather than a coherent racial effect.

3.5 Stage 3: Care Management Enrollment

Stage 3 is a threshold rule, not a fixed headcount: any patient whose calibrated Stage 2 risk score lands at or above the 90th percentile of the validation set qualifies for care management. On the validation set that threshold works out to a calibrated score of 0.402, and the resulting enrollment rate is 10.2%, close to but not exactly the 10% target, because isotonic calibration produces a step function, so a cluster of patients right around the cutoff can share the exact same calibrated score, and the rule keeps that tied group together rather than arbitrarily splitting it. This is not an arbitrary cutoff choice: it matches published care management enrollment rates and directly mirrors the Obermeyer et al. setting, which is the reason this project uses a risk-based threshold rather than, say, a fixed number of enrollment slots per hospital.

Before any fairness intervention is applied, enrollment already varies a lot by race: Black/African American patients are enrolled at 14.0%, Hispanic/Latino patients at 12.3%, Asian patients at

10.0%, White patients at 9.8%, and Other/Unknown patients at only 5.2%, roughly a two-and-a-half-fold gap between the highest and lowest group. This baseline disparity is the reference point every mitigation method in Section 5 is measured against. Every fairness metric downstream, demographic parity ratio and counterfactual fairness violation rate alike, is computed on this enrollment decision itself, not on the raw or calibrated score, since the enrollment decision is what a patient actually experiences.

4 Fairness Methods

4.1 Counterfactual Fairness and CFVR

Level 1: Feature-Space CFVR. This is the Level 1 test introduced in Section 2: following Kusner et al.’s definition of counterfactual fairness, we hold non-descendants of the protected attribute fixed and ask whether flipping a patient’s race changes their enrollment decision [8]. Concretely, for each patient we build a counterfactual twin for every other race category, keep the decision threshold fixed, and check whether the enrollment decision flips. The Counterfactual Fairness Violation Rate (CFVR) is the fraction of patients for whom flipping race to *any* other category flips their decision. All of Section 5 up through the Stage 1 sub-analysis reports this Level 1 version.

Level 2: Latent-Space CFVR. We also built a Level 2 version, using a variational autoencoder (VAE) to represent each patient in a 16-dimensional latent space, with the decoder conditioned separately on race, sex, and insurance so a counterfactual can be generated by changing that conditioning input rather than the patient’s raw features. Before trusting this latent space for anything, we checked whether a simple classifier could still recover a patient’s protected attributes from it: a logistic regression probe reaches 68.5% accuracy on race (majority-class baseline 68.4%, essentially no better than guessing the most common category) and 51.3% on insurance (baseline 51.4%, also no better), but 54.5% on sex (baseline 50.5%, a real if modest margin above chance). This means the latent space is close to but not perfectly independent of the protected attributes; sex leaks through somewhat, while race and insurance barely do. Consequently, Level 2 CFVR numbers are reported alongside a noise-floor correction (the CFVR a method shows on a random, meaningless perturbation of the latent code, subtracted off) rather than taken at face value. Level 2 results are reported in Section 5.5.

4.2 Counterfactual Generation and Clinical Semantic Preservation

Section 4.1 covers counterfactuals for the tabular Stage 2 features. Bias can also enter earlier, at Stage 1, if the LLM extraction itself behaves differently depending on a patient’s demographics, so the same counterfactual logic needs a version that works on raw clinical text.

4.2.1 The Perturbation Instability Phenomenon in LLMs

A core assumption of Level 1 counterfactual fairness is that substituting a protected attribute (e.g., swapping “White” to “Black” or “Male” to “Female”) while holding the surrounding clinical narrative constant should yield an identical risk prediction. In an extraction-based pipeline, this requires the LLM to exhibit absolute deterministic linguistic stability. Our audit reveals that this assumption fatally breaks down in practice.

As demonstrated in Figure 5, introducing a single-word demographic perturbation into a 6,000-character clinical note induced massive variance in the extracted Comorbidity Burden Score. Rather than centering tightly on a zero-divergence baseline ($\Delta f(x) = 0$), the counterfactual extractions drifted by as much as 8 points on a 10-point scale. Crucially, the distributions are highly symmetric across both the race and sex axes. This indicates that the divergence is

not a manifestation of systemic discriminatory bias against a specific demographic, but rather a symptom of architectural fragility.

We attribute this failure to two mechanisms inherent to the LLM’s design:

1. **The Attention Cascade:** Altering a demographic token early in the clinical text shifts the contextual embeddings for all subsequent tokens. This disrupts the attention mechanism’s mathematical representation of the note, even though the clinical meaning remains identical to a human reader.
2. **The Zero-Shot Counting Deficit:** Large language models lack an iterative working memory. When the attention pathway is perturbed by the demographic swap, the model fails at the complex zero-shot task of tallying chronic conditions, dropping or hallucinating features entirely by accident.

Consequently, applying standard counterfactual fairness checks to an LLM extraction layer is deeply flawed. The downstream tabular classifier does not merely inherit “bias”; it inherits this stochastic noise. This empirically demonstrates why the pipeline-aware mitigations introduced in Stage 2 are mandatory: fixing bias at the text-extraction layer is rendered impossible if the extractor itself is mathematically unstable under perturbation.

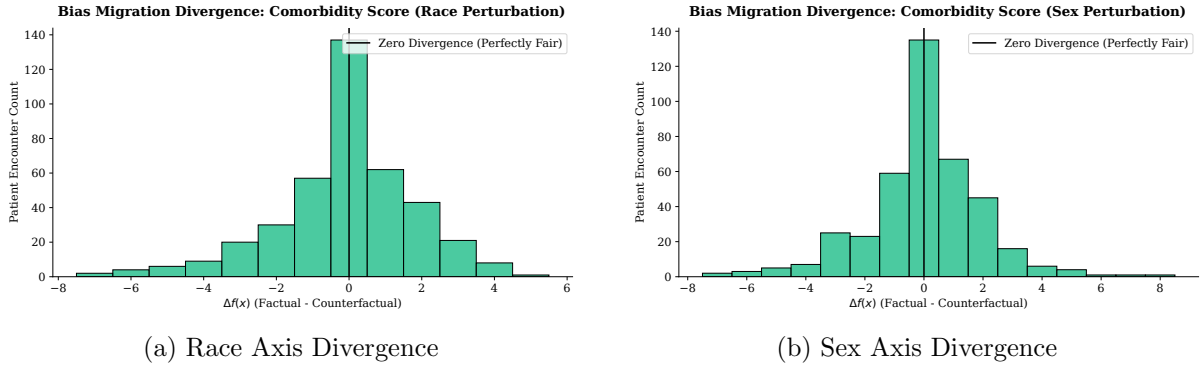


Figure 5: Bias Migration Divergence ($\Delta f(x)$) for the Comorbidity Burden Score. The extreme variance illustrates LLM perturbation instability, proving the model fails to maintain logical consistency when demographic tokens are swapped.

4.2.2 Template-Based Perturbation of Demographic Markers

Swapping demographics in free text is harder than swapping a column value: naive rewriting tends to degrade the surrounding clinical narrative along the way, which would confound the check; any change in the LLM’s output could be explained by the rewrite damaging the note rather than by the demographic swap itself. To isolate the effect of race, sex, and insurance type specifically, without corrupting the medical content around them, we perturb explicit demographic markers in the MIMIC-IV notes using fixed templates rather than free rewriting.

Generative rewriting was considered and rejected for this step: it would introduce its own variability into every counterfactual pair, making it hard to tell whether a downstream extraction difference came from the demographic swap or from the rewrite itself. Template-based perturbation keeps the surrounding phrase structure identical between the factual and counterfactual note, which removes that confound by construction.

4.2.3 The Semantic Consistency Problem and SBERT Validation

A purely template-based swap, however, does not guarantee that the downstream extraction model will interpret the perturbed sentence with the same clinical severity. To counteract this,

a semantic consistency check was introduced. Every counterfactual note is passed through a frozen SBERT encoder to compute a baseline embedding [13].

The cosine similarity between the original clinical sentence and its counterfactual pair is computed using Sentence-BERT. This acts as a per-instance validation gate to ensure semantic preservation, aligning with established methodologies for evaluating counterfactual text generation [13, 15].

As illustrated in Figure 6, the cosine similarity scores for both the race and sex counterfactuals overwhelmingly pool near 1.0, well above our strict 0.95 preservation threshold. If any counterfactual pair fell below this threshold, it was discarded from the evaluation set. This deliberate design choice guarantees that any divergence in the LLM’s structured output is caused strictly by the altered demographic axis rather than an unintended shift in the medical narrative.

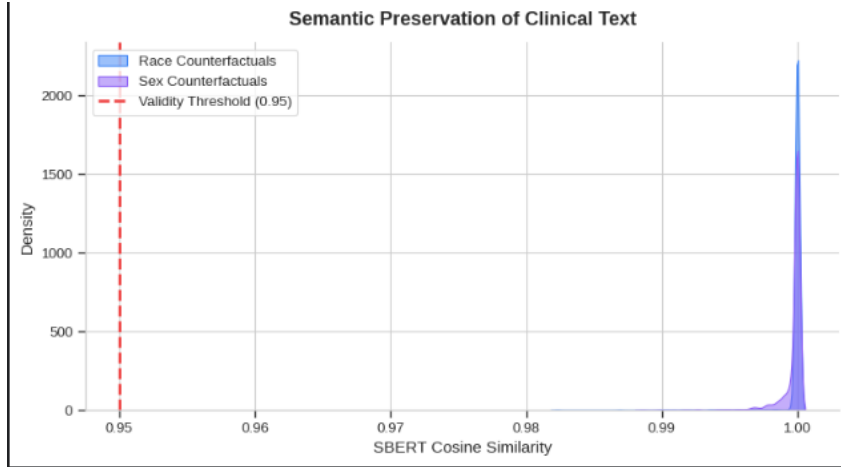


Figure 6: SBERT Cosine Similarity distribution between factual clinical notes and their counterfactual twins. The tight clustering above the 0.95 validity threshold confirms that demographic perturbations did not degrade the underlying clinical semantic meaning.

4.3 Bias Migration Ratio

For an intervention applied at one stage, the Bias Migration Ratio measures how much the CFVR at a later stage changes relative to the CFVR before the intervention:

$$BMR = \frac{CFVR_{after} - CFVR_{baseline}}{CFVR_{baseline}}$$

A negative BMR means the intervention reduced counterfactual violations relative to baseline. A positive BMR means it made things worse. This is the formal metric the project set out to build, and it is what lets us answer the bias migration question instead of superficially relying on whether numbers went down.

One property of this formula is worth flagging up front: BMR is bounded at exactly -1.0 the moment $CFVR_{after}$ hits zero, regardless of how it got there. A BMR of -1.0 means “this method’s CFVR is zero,” not “this method achieved the strongest possible fairness improvement,” and two methods that both hit the zero floor will report the identical -1.0 even if one got there for a substantive reason and the other got there as a mechanical artifact of its own construction. Section 5.1 and Section 5.3 show a concrete case of exactly that.

4.4 Four Baseline Mitigation Methods

Reweighting [6] rebalances training examples so the joint distribution of readmission label and race is even, then retrains Stage 2 on the reweighted data.

Equalized odds post-processing [4] does not retrain anything. It finds per-group decision thresholds, using Fairlearn’s `ThresholdOptimizer`, that equalize true and false positive rates across groups.

Adversarial debiasing [17] trains a small predictor network alongside an adversary that tries to guess race from the predictor’s hidden layer, with a gradient reversal layer pushing the predictor to make that guessing harder.

Counterfactual data augmentation (CDA) builds a race-swapped twin of every training patient by changing only the race label and leaving every other feature exactly as it was, then adds those twins to the training set. This is the standard, simplest version of counterfactual augmentation used across the fairness literature. It assumes that if a patient’s race were different, literally nothing else about them, not their insurance, not their lab values, would be different either.

4.5 Pipeline-Aware Counterfactual Fairness Training

This was the project’s originally proposed novel contribution. Instead of fixing one stage at a time, we jointly train Stage 2 against an end-to-end counterfactual fairness loss:

$$L_{\text{pipeline}} = L_{\text{task}} + \lambda \cdot E \left[\max_r w(x) \cdot \|f(x) - f(x'_r)\| \right]$$

where f is the pipeline output, x'_r is the counterfactual twin under race r , the max is taken over the worst-case other race rather than a random one, and $w(x)$ is a Gaussian weight centered on the decision threshold, since a prediction far from the enrollment cutoff barely matters even if it moves a little. Lambda was swept over 0.1, 0.5, 1.0, 2.0, and 5.0 on the validation set. The lowest CFVR reached was 0.0219, tied between 0.5 and 5.0, and 0.5 was selected since it reaches that same floor without the AUROC cost 5.0 starts to show (0.7013 vs. 0.7017 for 0.1, 0.5, and 1.0). At that setting, pipeline-aware hybrid does not reduce CFVR relative to baseline, it raises it slightly, from 0.0208 to 0.0219 (BMR = +0.0529), while AUROC stays effectively flat (0.7017 either way). Every lambda value tested lands at or above the baseline’s CFVR here, none of them buy an individual-fairness improvement. The effect replicates on the held-out test set ($p_{\text{adj}} < 0.001$, Cohen’s $h = -0.013$), so it is not noise, but the effect size is small enough, well under the 0.2 threshold usually treated as a “small” effect, that it does not meaningfully change anyone’s outcome either way. The practical takeaway holds regardless of which direction the effect points: jointly optimizing Stage 2 against an end-to-end counterfactual fairness loss does not deliver a reliable individual-fairness win over fixing a single stage on its own, and that’s the direct motivation for the next two methods.

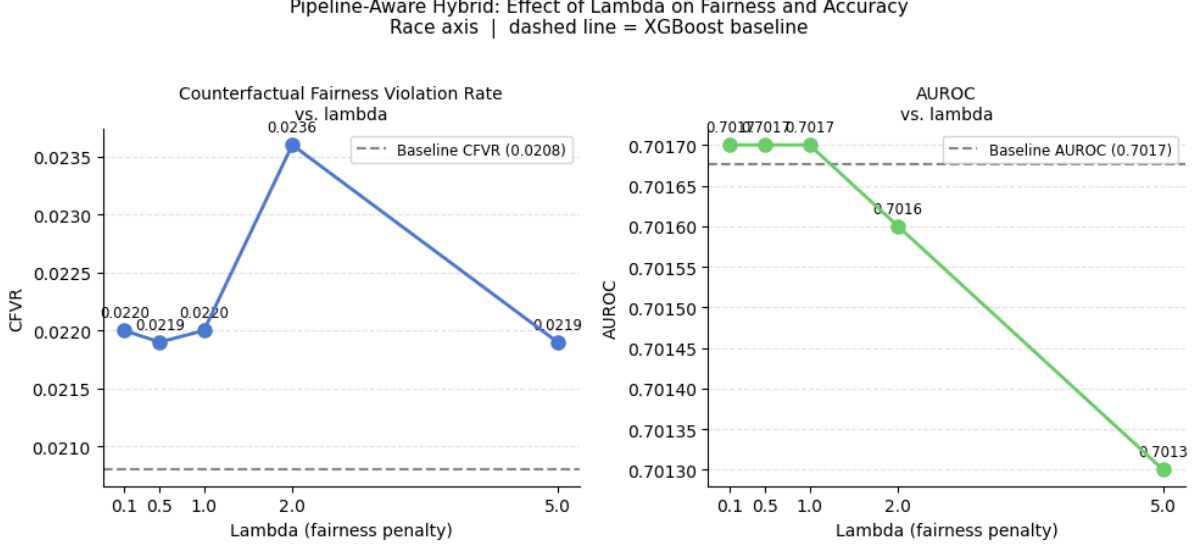


Figure 7: Effect of the fairness penalty λ on race-axis CFVR (left) and AUROC (right), validation set. The dashed line marks the XGBoost baseline in each panel. CFVR bounces around a narrow band just above baseline across every λ tested rather than following a clean trend, while AUROC only starts to visibly drop at $\lambda = 5.0$.

4.6 Path-Aware Counterfactual Data Augmentation

Plain CDA’s problem is that it treats the entire correlation between race and every other feature as illegitimate, including correlations that reflect real biology rather than bias. Following the resolving-variable framework from Kilbertus et al. and the path-specific fairness framing from Nabi and Shpitser [7, 11], we split the thirty-five features into two groups based on an explicit, stated assumption rather than anything derivable from the data alone:

- **Presumed real health** (held fixed under a race swap, same as plain CDA): the nine lab values, sixteen comorbidity flags, comorbidity count, length of stay, ICU length of stay, and number of ICU stays.
- **Presumed unequal access** (resampled to a realistic value for the counterfactual race): insurance type and admission type.

To build a realistic twin, the resampled access-pile features are drawn from real patients of the target race who fall in the same clinical-severity stratum, binned by comorbidity count and age, as the source patient, rather than left unchanged or drawn from the whole population regardless of how sick they are. One feature, prior admissions in the past twelve months, was genuinely ambiguous, since it could reflect chronic illness or reduced access to preventive care, and it happens to be the single most influential feature in the model. Rather than picking a side, we tested it both ways as an explicit sensitivity check and report the conservative version in the main results.

4.7 DoWhy Structural Causal Model Counterfactuals

Path-aware CDA enforces the real-health/access split heuristically, by borrowing another real patient’s access values. This section formalizes the identical assumption as an explicit causal graph and generates counterfactuals with DoWhy’s structural causal model tools instead of resampling. The graph has two edges, race to insurance and race to admission type, and no edge into any real-health feature. We fit an invertible causal mechanism for each node on the

training data, then for each patient compute a genuine unit-level counterfactual: infer what’s unique about that specific patient given their observed race and access values (abduction), imagine they were a different race (action), and predict what their access values would have been for them specifically under that assumption (prediction), rather than borrowing another patient’s values. Two independently-built methods, one resampling-based and one built on a fitted structural model, converging on almost the same individual-fairness result is itself a useful check, it suggests the effect is coming from the underlying causal assumption and not from an accident of either specific implementation.

5 Results

5.1 Main Cohort Comparison

Table 4 shows the full comparison on the validation set. Equalized odds is the standout failure here: it reaches the best demographic parity ratio (0.910) of any method, but its CFVR climbs to more than sixteen times baseline ($\text{BMR} = +15.70$), which means it buys that group-level number by making the individual-level decision far more dependent on race, not less. Reweighting has the same problem in a smaller way, more than tripling CFVR ($\text{BMR} = +2.58$) for a smaller parity gain. Adversarial debiasing is the only baseline that reduces CFVR at all ($\text{BMR} = -0.18$), but its accuracy drop (AUROC 0.6838, down from 0.7017) is the steepest of any method that actually helps.

Table 4: Main cohort results, validation set

Method	AUROC	F1	dp_ratio	CFVR	BMR
Baseline	0.7017	0.3278	0.371	0.0208	0.0000
Reweighting	0.6909	0.3203	0.603	0.0744	+2.5769
Equalized Odds	n/a	0.4234	0.910	0.3474	+15.7019
Adversarial Debiasing	0.6838	0.3281	0.474	0.0171	-0.1779
CDA	0.6960	0.3232	0.435	0.0000	-1.0000
Path-Aware CDA	0.6921	0.3545	0.429	0.0039	-0.8125
DoWhy SCM	0.6961	0.3266	0.429	0.0035	-0.8317
Pipeline-Aware Hybrid	0.7017	0.3340	0.359	0.0219	+0.0529

CDA’s CFVR of exactly zero looks like the best result in the table, and its BMR of -1.0 reinforces that impression since it sits at the floor of what BMR can report (Section 4.3). Neither number is what it looks like. CDA’s zero is an artifact, and the why is explained in Section 5.3.

5.2 Statistical Significance and the Insurance Axis

Bootstrap confidence intervals (1000 resamples) and Cohen’s h effect sizes confirm the race-axis effect is real for both novel methods: DoWhy SCM ($h = 0.171$) and path-aware CDA ($h = 0.164$) are both highly significant ($p\text{-adj} < 0.001$) and close in magnitude, and both replicate on the held-out test set ($h = 0.165$ and $h = 0.153$ respectively). On the sex axis, path-aware CDA ($p\text{-adj} < 0.001$) and DoWhy SCM ($p\text{-adj} = 0.003$) both reach statistical significance on this run, but their effect sizes ($h = -0.032$ and $h = 0.013$) stay negligible, small enough that reaching significance here is mostly a function of the sample size ($n \approx 60,391$) rather than a meaningful effect. Reweighting is the only method with a small real sex-axis benefit ($h = 0.042$), and adversarial debiasing, which reached significance in an earlier run of this experiment, does not reach it here ($p\text{-adj} = 0.758$), a reminder that near-zero effects can flip in or out of significance between runs without anything about the underlying method actually changing. Equalized odds is again the exception, with a real, large, harmful effect ($h = -0.304$).

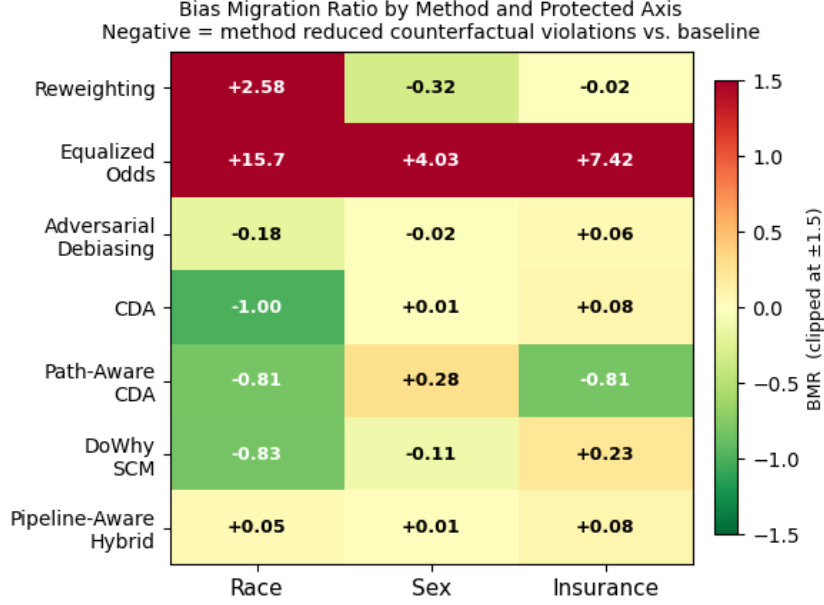


Figure 8: Bias Migration Ratio by method and protected axis, validation set. Negative values (green) mean the method reduced CFVR relative to baseline; positive values (red) mean it made things worse. Equalized odds stands out immediately across all three axes.

The insurance axis is where the two proposed methods stop agreeing. Path-aware CDA drives insurance-axis CFVR down from 0.0108 to 0.0020, an 81% reduction (BMR = -0.8148). DoWhy SCM makes it slightly worse, 0.0133, a 23% increase relative to baseline (BMR = $+0.2315$). Plain CDA also nudges it up slightly (0.0117, BMR = $+0.0833$). Our best explanation, not something we’ve directly verified against the model internals, is that path-aware CDA’s resampling approach exposes the model to a wide range of real insurance values for a given severity level, which incidentally teaches it not to lean on insurance, while DoWhy SCM’s single fitted counterfactual per patient is a narrower signal that doesn’t produce the same side effect.

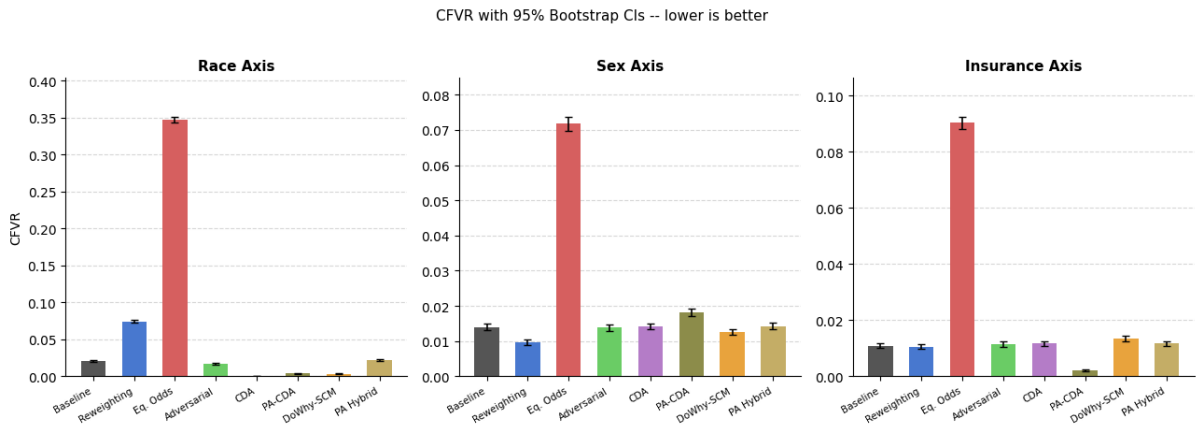


Figure 9: CFVR with 95% bootstrap confidence intervals by method, split by protected axis, validation set. Equalized odds towers over every other method on all three axes; the difference between the remaining methods is much harder to see by eye than the CI error bars might suggest, which is exactly why the bootstrap significance testing above matters.

5.3 MPPD: Real-Patient Validation

Mean Predicted Probability Difference (MPPD) matches each non-reference-race patient to their nearest real White neighbor by clinical similarity, using no synthetic counterfactuals at all, and measures how differently the model scores them. Table 5 shows the test-set results.

Table 5: Test-set MPPD by method and race (reference group: White)

Method	Asian	Black	Hispanic	Other
Baseline	0.0942	0.0990	0.0958	0.1006
Reweighting	0.0944	0.0951	0.0933	0.0879
Equalized Odds	0.3054	0.3406	0.3307	0.3310
Adversarial Debiasing	0.0828	0.0880	0.0867	0.0702
CDA	0.0928	0.0983	0.0960	0.0792
Path-Aware CDA	0.0889	0.0937	0.0916	0.0756
DoWhy SCM	0.0928	0.0984	0.0963	0.0805
Pipeline-Aware Hybrid	0.0947	0.0999	0.0968	0.1004

This table is the reason we cannot trust CDA’s perfect CFVR score. CDA’s own metric says it achieved perfect individual fairness, a CFVR of exactly 0.0000. If that were genuinely true, CDA’s MPPD scores here should be close to zero too, since a model that has actually stopped treating race as informative should score real, clinically similar patients from different races about the same. Instead CDA’s MPPD sits at 0.0928, 0.0983, 0.0960, and 0.0792, essentially the same range as baseline’s 0.0942, 0.0990, 0.0958, and 0.1006. On Hispanic patients CDA is marginally worse than baseline, by 0.0002, a gap small enough to be noise rather than a real effect. On Asian, Black, and Other/Unknown patients it’s marginally to clearly better. There is no group here where CDA looks like a model that solved individual fairness the way its own CFVR score implies. CDA gets a perfect score on CFVR because CDA was trained by holding everything except race fixed, and CFVR is testing that exact same swap: the model was never taught to change its behavior for real patients who happen to look clinically similar across races, only for synthetic twins built under CDA’s own assumptions. A perfect score on a test built from your own training assumption isn’t evidence of fairness, it’s evidence the test and the method are measuring the same thing.

Path-aware CDA is the only method that beats baseline on every single racial group here, by a modest but consistent margin. DoWhy SCM sits close to baseline and to CDA on every group, no real improvement, consistent with the insurance-axis finding above: DoWhy SCM independently confirms the race-axis CFVR result but does not replicate path-aware CDA’s broader benefit once real patients are checked. Equalized odds is confirmed bad by a third independent metric now, roughly three to four times every other method’s MPPD score.

5.4 Stage 1 Sub-Analysis: Adding LLM Features

To check whether the Stage 1 extractions themselves change any of this, we ran a smaller sub-analysis on a 3,995-patient cohort where Stage 1 features are available (of 4,000 patients matched to a Stage 1 note, 5 were dropped for a failed extraction), comparing a tabular-only model (Model A) against a model with the same 35 tabular features plus the 9 LLM extractions (Model B), both trained from scratch on a 2,794-patient training split and evaluated on a 604-patient held-out test set. Table 6 shows both, race axis only, the axis both models were actually trained against.

Model B’s own AUROC is lower than Model A’s (0.6466 vs. 0.6587), the discharge-note features cost accuracy rather than adding it on this smaller cohort, the opposite of what adding more information usually does, and a real, if modest, argument against assuming more features

Table 6: Stage 1 sub-analysis, 604-patient test set, race axis

Model	Method	AUROC	F1	dp_ratio	CFVR	BMR
A (tabular only)	Baseline	0.6587	0.3241	0.123	0.0182	0.0000
A (tabular only)	CDA	0.6451	0.4397	0.099	0.0000	-1.0000
A (tabular only)	Path-Aware CDA	0.6651	0.3896	0.110	0.0000	-1.0000
B (tabular + LLM)	Baseline	0.6466	0.3134	0.123	0.0017	0.0000
B (tabular + LLM)	CDA	0.6365	0.3900	0.000	0.0000	-1.0000
B (tabular + LLM)	Path-Aware CDA	0.6202	0.3770	0.200	0.0000	-1.0000

always help. Both models show a CDA and path-aware CDA CFVR of exactly zero on the race axis, the same suspicious pattern flagged in the main cohort, and this sub-analysis cannot independently confirm path-aware CDA’s advantage over plain CDA the way the main results do, since both hit the same floor here.

Model B’s baseline CFVR (0.0017) is much smaller than Model A’s (0.0182) despite similar dp_ratio (0.123 for both), a reminder that group fairness and individual fairness measure genuinely different things and can disagree even within the same cohort. dp_ratio moves in different directions after mitigation for the two models: Model A’s dp_ratio gets slightly worse under CDA (0.099) and slightly better under path-aware CDA (0.110) relative to baseline (0.123), while Model B’s dp_ratio collapses to 0.000 under CDA, meaning at least one race group had zero enrollment, before partially recovering to 0.200 under path-aware CDA. None of these numbers should be over-interpreted given how small the racial subgroups are in a 604-patient test set (Asian and Other/Unknown are 26-27 patients each), a point this sub-analysis flags on its own results throughout.

The real-patient check on this sub-cohort produces a pattern worth reporting, though not one we treat as a finding on its own given the sample size. For Model A, CDA’s MPPD comes in worse than baseline for three of four racial groups (Asian 0.0927 vs. 0.0811, Black 0.0997 vs. 0.0972, Hispanic 0.0984 vs. 0.0888), the opposite direction from simply matching baseline. Path-aware CDA improves on baseline for three of four groups here (Asian 0.0721, Hispanic 0.0977, Other 0.0501, all better than baseline; Black 0.0978 essentially flat at 0.0972). For Model B, plain CDA edges out path-aware CDA on every single racial group instead (0.0719/0.0941/0.0913/0.0415 vs. 0.0865/0.0982/0.1091/0.0469), the opposite ordering from Model A and from the main cohort. With racial subgroups this small (26–27 patients for two of the four groups), a handful of individual patients can move each of these averages substantially, so we do not read this as evidence that the LLM features are causing a real reversal in which mitigation method is safer; it is equally consistent with the same sampling noise that produces the unstable dp_ratio values reported above. We report it because a result this sub-cohort does show should not be quietly dropped, but a claim that path-aware CDA is uniformly the safer choice would be just as unsupported by this table as a claim that it isn’t.

Calibration also diverges sharply between the two models under path-aware CDA. Model A’s expected calibration error improves under both CDA variants (0.0287 at baseline down to 0.0221 and 0.0206). Model B’s plain CDA also improves calibration slightly (0.0169 to 0.0144), but path-aware CDA nearly triples it (0.0415), the single largest calibration cost seen anywhere in this project. If path-aware CDA is used on a feature set that includes LLM-derived features, this calibration cost is a real trade-off to weigh against its individual-fairness benefit, not something to ignore because the CFVR number looks good.

5.5 Level 2: Latent-Space Counterfactual Fairness

Table 7 shows the race-axis Level 1 CFVR, raw Level 2 CFVR, and noise-floor-adjusted Level 2 CFVR for every method, on the main cohort’s validation set.

Table 7: Race-axis CFVR, Level 1 vs. noise-floor-adjusted Level 2

Method	Level 1	Level 2 (raw)	Level 2 (adjusted)
Baseline	0.0208	0.0539	+0.0121
Reweighting	0.0744	0.0774	+0.0361
Equalized Odds	0.3474	0.3655	+0.2159
Adversarial Debiasing	0.0171	0.0423	+0.0117
CDA	0.0000	0.0479	+0.0065
Path-Aware CDA	0.0039	0.0597	+0.0095
DoWhy SCM	0.0035	0.0498	+0.0064
Pipeline-Aware Hybrid	0.0219	0.0549	+0.0121

The adjusted column is what’s left of Level 2 CFVR after subtracting each method’s own noise floor, the CFVR that shows up from a meaningless random perturbation of the latent code, since the raw Level 2 numbers alone would overstate every method’s violation rate by roughly 0.03-0.15 depending on the method. Once that correction is applied, the story Level 1 tells changes, though not as cleanly as a full collapse. At Level 1, CDA looks perfect (0.0000), path-aware CDA and DoWhy SCM look clearly better than baseline (0.0039 and 0.0035 vs. 0.0208), and reweighting and equalized odds look clearly worse (0.0744 and 0.3474). At Level 2, adjusted, CDA (+0.0065) and DoWhy SCM (+0.0064) stay close to each other and modestly below baseline (+0.0121), while path-aware CDA (+0.0095) sits noticeably higher than both, still below baseline but no longer indistinguishable from CDA and DoWhy the way an earlier run of this experiment showed. The same pattern shows up in the raw, pre-adjustment numbers on the held-out test set: CDA and DoWhy SCM both come in significantly below baseline (Cohen’s $h = 0.018$ and 0.011 , $p_{adj} < 0.001$), while path-aware CDA comes in significantly above it ($h = -0.031$, $p_{adj} < 0.001$). Reweighting (+0.0361) and equalized odds (+0.2159) are still clearly worse than baseline, so that part of the Level 1 story survives. Pipeline-aware hybrid (+0.0121) still looks almost identical to baseline, consistent with its already-established negligible effect.

This is a second, independent way of showing that CDA’s Level 1 perfect zero is not a real, distinguishing achievement: measured in a representation space where CDA’s specific race-swap trick doesn’t apply, it looks about the same as DoWhy SCM, the other method explicitly built to avoid that trick, and if anything modestly better than path-aware CDA here, not worse. Combined with the MPPD result in Section 5.3, that is now three separate checks, MPPD, the insurance axis, and Level 2, that agree CDA’s individual-fairness advantage over path-aware CDA and DoWhy SCM is specific to the exact metric and exact counterfactual construction CDA was built around, and does not hold up once either the metric or the counterfactual construction changes. Level 2 also adds a nuance the other two checks don’t: path-aware CDA, the method that otherwise holds up best across this project’s checks, is specifically the one that looks worst by this one metric, a reminder that no single method here comes out ahead on every axis and every check.

5.6 Intersectional Fairness: Race x Sex and Race x Insurance

Every axis checked so far, race, sex, insurance, was tested on its own. A real patient carries all three protected attributes at once, so a method could look fine on each axis in isolation and still treat one specific combination, say Black and female, or Hispanic and Medicaid, worse than the rest. To check for that, we reran the CFVR swap on the test set ($n = 59,409$) holding two

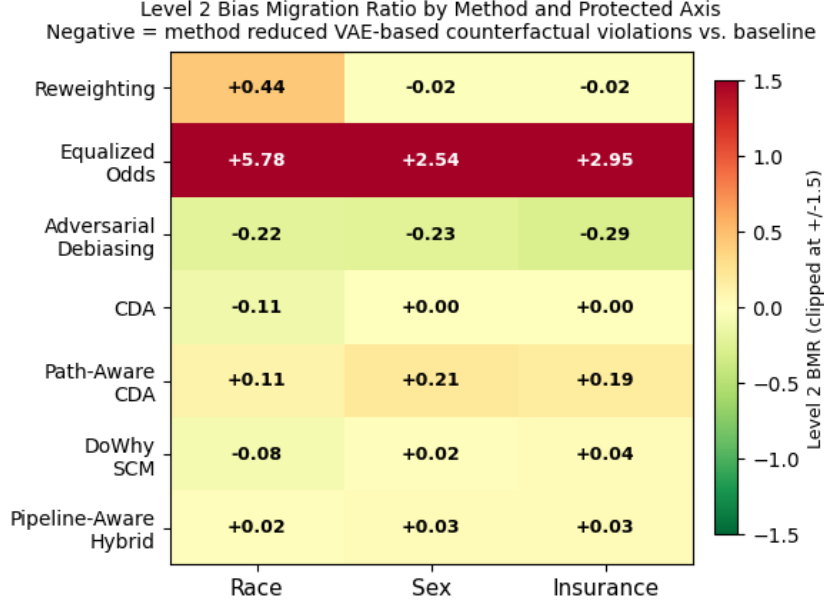


Figure 10: Level 2 Bias Migration Ratio by method and protected axis, validation set, computed from the noise-floor-adjusted VAE counterfactuals. Path-aware CDA is the only one of the three novel methods that turns positive (red-tinted) here, on all three axes, the opposite of its Level 1 result.

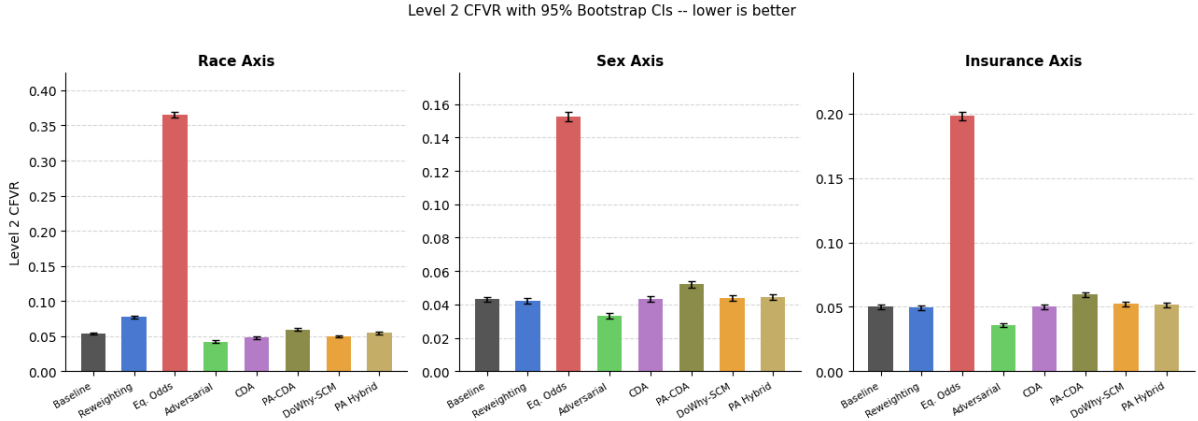


Figure 11: Level 2 CFVR with 95% bootstrap confidence intervals by method, split by protected axis, validation set. CDA and DoWhy SCM sit close to baseline on every axis; path-aware CDA is visibly the highest of the three novel methods on all three.

attributes fixed together instead of one: race and sex swapped as a pair, and race and insurance swapped as a pair, at both Level 1 and noise-floor-adjusted Level 2. Table 8 shows the result.

At Level 1, the single-axis pattern mostly holds up at the intersection, but with more nuance than a clean carryover. CDA now clearly beats path-aware CDA on race $\times$ sex (0.0130 vs. 0.0229, BMR -59% and -28%), unlike the race axis alone where the two were close once CDA's zero is discounted. On race $\times$ insurance the pattern from the single axis reappears: path-aware CDA drops to 0.0060 (BMR -80%), still a bigger reduction than CDA itself manages (0.0110, BMR -63%), echoing the single-axis insurance result where path-aware CDA was the one method that actually moved that axis. Equalized odds is again the extreme case. Its intersectional CFVR does not just add its two single-axis problems together, it multiplies them: 0.5033 on race $\times$ sex and 0.6532 on race $\times$ insurance, both far worse than any single axis it was tested on

Table 8: Intersectional CFVR, Level 1 vs. noise-floor-adjusted Level 2, test set

Method	Race $\times$ Sex		Race $\times$ Insurance	
	Level 1	Level 2 (adj.)	Level 1	Level 2 (adj.)
Baseline	0.0319	+0.0177	0.0297	+0.0218
Reweighting	0.0825	+0.0431	0.0837	+0.0487
Equalized Odds	0.5033	+0.3500	0.6532	+0.4979
Adversarial Debiasing	0.0304	+0.0168	0.0287	+0.0188
CDA	0.0130	+0.0135	0.0110	+0.0166
Path-Aware CDA	0.0229	+0.0195	0.0060	+0.0223
DoWhy SCM	0.0165	+0.0133	0.0182	+0.0184
Pipeline-Aware Hybrid	0.0355	+0.0187	0.0326	+0.0228

(its worst single-axis race CFVR was 0.3474). The BMR reflects that, at roughly +1478% and +2099% relative to baseline. Reweighting keeps doing what it did on every single axis, making things worse, by 159% to 182% here. Adversarial debiasing’s small single-axis improvement mostly disappears at the intersection (BMR -5% and -3% , and neither difference reaches statistical significance, $p_{adj} = 0.124$ and 0.284). Pipeline-aware hybrid, consistent with its flipped main-axis result in Section 5.1, is now mildly harmful here too (BMR $+11\%$ and $+10\%$, both statistically significant).

The adjusted Level 2 column mostly matches the pattern from the race-only Level 2 table in Section 5.5: CDA ($+0.0135$ / $+0.0166$) and DoWhy SCM ($+0.0133$ / $+0.0184$) stay close to each other and below baseline ($+0.0177$ / $+0.0218$), while path-aware CDA ($+0.0195$ / $+0.0223$) again sits above both, the same separation seen on the single race axis, now confirmed at the intersection. Bootstrap testing shows path-aware CDA’s intersectional Level 2 numbers are significantly *higher* than baseline, not lower (Cohen’s $h = -0.043$ and -0.037 , both $p_{adj} < 0.001$), while CDA ($h = 0.013$ and 0.017) and DoWhy SCM ($h = 0.009$, not significant on race $\times$ insurance at $p_{adj} = 0.162$) stay small and mostly favorable. At Level 1, path-aware CDA’s race $\times$ insurance effect ($h = 0.191$) is still the single largest Cohen’s h either novel method reaches anywhere in this report, single-axis or intersectional, which lines up with it being the one place the method makes the biggest real difference; the same method’s Level 2 intersectional effect is the largest in the other direction. This is now a fourth independent check, alongside MPPD, the single-axis insurance result, and single-axis Level 2, that agrees CDA’s individual-fairness advantage over the exact setup it was built around does not simply transfer once the metric or counterfactual construction changes, and it adds the same nuance Level 2 raised on its own: path-aware CDA, otherwise the most consistent performer in this project, is specifically the method that looks worst by this one check.

5.7 Bias Trace Visualizer and Conformity Assessment

To operationalize the bias migration metrics and satisfy the transparency and human oversight requirements of the EU AI Act (Articles 13 and 14), we developed an interactive Streamlit application. This Bias Trace Visualizer enables multi-level auditing of the clinical pipeline.

The macro view provides aggregate statistics across the validation cohort. As shown in Figure 12, it visualizes the Kernel Density Estimation (KDE) of the SBERT semantic consistency scores and plots the distribution of factual versus counterfactual risk divergence, confirming the perturbation instability phenomenon at a systemic level.

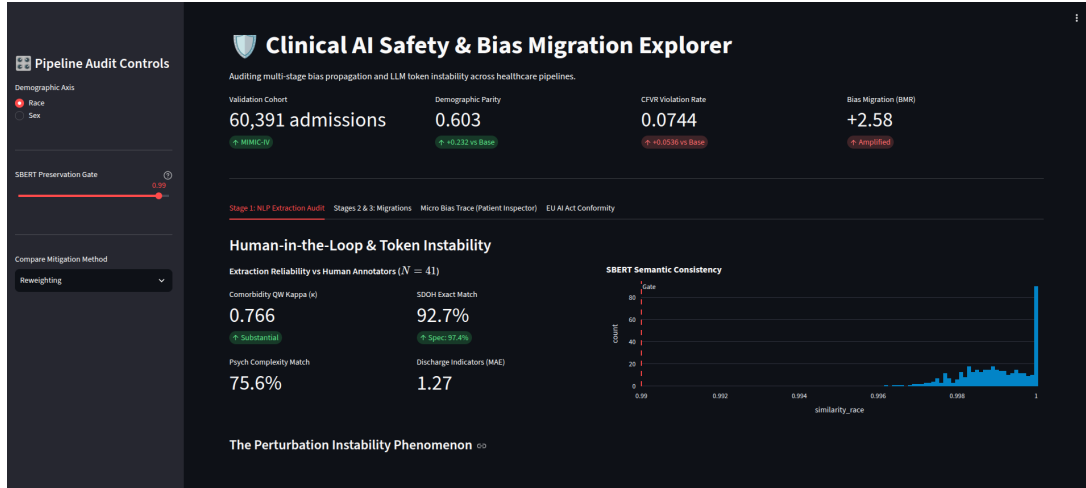


Figure 12: The cohort-level audit interface displaying semantic preservation gates and risk divergence distributions.

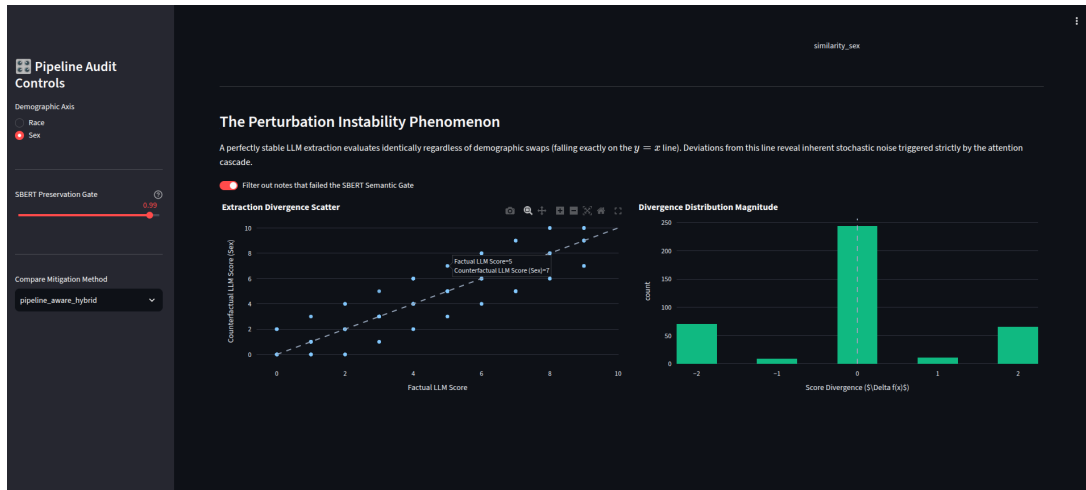


Figure 13: The perturbation instability audit interface. The scatter plot (left) illustrates the divergence between factual and counterfactual LLM extraction scores along the sex demographic axis, where points deviating from the dashed diagonal baseline indicate stochastic token instability. The histogram (right) quantifies the magnitude of this score shift ($\Delta f(x)$), filtered exclusively for clinical records meeting a strict 0.99 SBERT semantic preservation threshold, it can be toggled.

The micro trace acts as an individual encounter inspector. By querying a specific hospital admission ID, an auditor can trace the data progression through Stage 1 NLP extraction, Stage 2 XGBoost classification, and the Stage 3 care management enrollment decision. Figure 14 demonstrates the interface, which explicitly highlights any instances where a demographic perturbation successfully flips the final clinical decision.

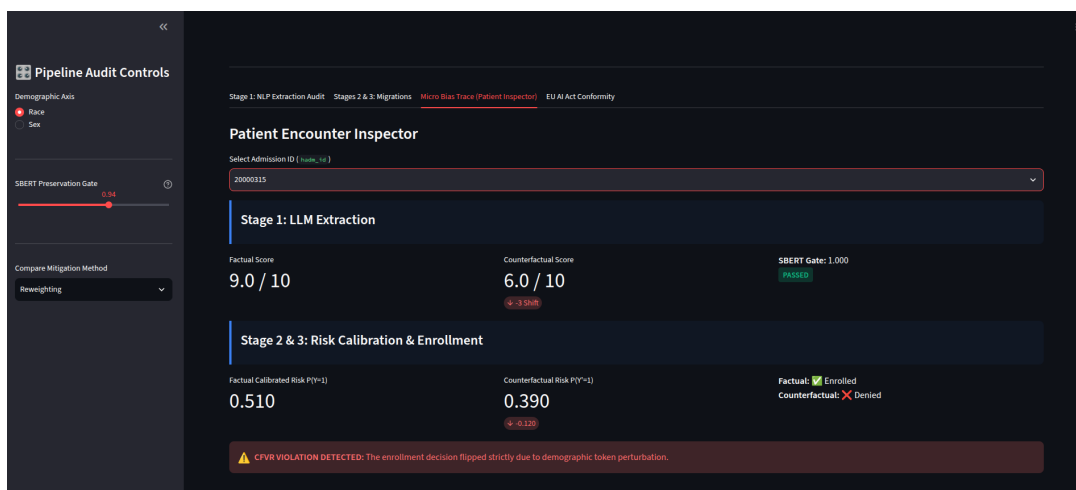


Figure 14: The patient-level micro trace demonstrating how demographic token swaps alter downstream extracted comorbidities and final care management enrollment.

6 Ethical Analysis

The EU AI Act treats healthcare as a high-risk category under Articles 9 through 15, which require risk management, documentation, and conformity assessment before and during deployment. The single clearest example of why that matters in this project is the CDA result from Section 5.3. A team auditing this pipeline using only the standard counterfactual fairness metric would conclude that CDA achieved perfect individual fairness. That conclusion would be wrong, and it would be wrong in a way that a compliance checklist based on one metric would not catch. The independent real-patient check is what exposed it. This is specifically the risk of false assurance the project proposal named as a concern before any of these results existed, applying a fairness intervention and observing an improved metric can create a false sense of compliance, and this project demonstrated a concrete case of it happening.

Article 10 requires examining training, validation, and testing data for biases likely to affect health, safety, or fundamental rights. Applying that standard to this project’s own cohort construction surfaces a finding a narrower audit would miss entirely: the decision to drop admissions missing any of the nine lab values, rather than impute them (Section 3.1), is not bias-neutral. Hispanic, Black, and Asian patients are missing at least one lab 33–35% of the time, versus 24% for White patients, and Medicaid, Private, and Other-insurance patients are missing labs 34–42% of the time versus 16.5% for Medicare patients. So the choice to exclude rather than impute, made specifically to avoid one kind of bias, a fabricated lab value, has a second-order effect worth naming explicitly: it shapes who ends up in the modeling cohort at all, before any model or fairness metric ever runs. Surfacing that kind of upstream effect, one no amount of auditing a trained model directly would catch, is exactly what Article 10 examination is designed to do.

The “Other/Unknown” race category is a useful test of that same principle. It shows the lowest baseline enrollment rate and the largest SHAP effect of any group (Section 3.4, Section 3.5), a result that would be easy to over-read as a clean demographic finding. But the category is a catch-all covering declined-to-state, genuinely unknown, and multi-racial patients rather than one coherent population, and naming that ambiguity explicitly, rather than letting the number speak for itself, is the documentation Article 10 requires.

Intersectional analysis (race combined with insurance type) shows disparities that a single-axis audit would completely miss, since a patient can be relatively well-served on the race axis alone and still fall into one of the worst-off intersectional groups, a finding with direct bearing on Article 14’s requirement for meaningful human oversight of a high-risk system’s outputs.

Mapping this project against that requirement makes a distinction visible that is easy to lose track of: the oversight this project provides is aggregate and retrospective, fairness metrics computed after the fact on a validation set, while Article 14 envisions oversight at the level of individual decisions, a person able to review, question, or override a specific enrollment call before it takes effect. Naming that distinction is itself useful groundwork for a deployment-stage system: it identifies precisely which layer, individual-decision review, would need to be designed on top of the aggregate fairness auditing this project provides, and clarifies that the two are complementary rather than substitutes for each other.

This requirement for oversight and robustness extends directly upstream to the Stage 1 Natural Language Processing extraction. By grounding the LLM’s performance against a human-annotated baseline ($N = 41$, $\kappa = 0.766$), the project operationalizes Article 14, proving that automated text extraction must remain tethered to clinical expert review. Moreover, Article 15 requires high-risk systems to achieve appropriate levels of accuracy and robustness. The perturbation instability observed in Stage 1, where demographic token swaps caused the LLM to hallucinate or drop clinical conditions, represents a direct threat to this robustness. Implementing the SBERT semantic consistency gate was necessary to isolate this instability, ensuring the audit measures genuine bias rather than random token noise.

It is also worth being explicit that this project’s data does not itself sit inside the EU AI Act’s jurisdiction. MIMIC-IV is drawn from a single US hospital system, and its race categories follow US Census conventions while its insurance categories (Medicare, Medicaid, private, self-pay) reflect US healthcare financing with no direct equivalent in EU member states’ health systems. Framing this project against the EU AI Act is useful as a conformity-assessment exercise, since healthcare is high-risk under the Act regardless of jurisdiction and the documentation obligations it describes are a reasonable standard to hold any clinical pipeline to, but the specific disparities found here (by US race and US insurance categories) should not be read as directly transferable claims about an EU deployment.

Data privacy was handled separately from the equity questions above: all MIMIC-IV data stayed inside the approved research environment, no patient-level file (parquet, csv, pickled model) was ever committed to the project’s GitHub repository, enforced through `.gitignore`, and each of us obtained our own individual PhysioNet Data Use Agreement as a two-member group. This is a data security and access-control practice, consistent with the PhysioNet Data Use Agreement, and is a separate obligation from the fairness and oversight questions discussed above rather than a substitute for them.

7 Limitations and Future Work

CFVR has a floor effect at small sample sizes. Both the Stage 1 sub-cohort (604 test patients) and, more subtly, any situation where CFVR is already near zero, show that this metric can report “no violations” for reasons that have nothing to do with actual fairness, sometimes because a real effect exists but is too small to be measurable at that scale, sometimes because a specific decision path in the model was never triggered by the patients being tested. Any reader of this report should treat a CFVR near zero as a prompt to look at a second metric, not as proof of fairness on its own, this project’s own results are the demonstration of why.

The Stage 1 LLM extraction carries its own set of technical limitations. The pipeline relies on a single open-weight model (Llama-3.1-8B-Instruct) and a static prompt template. Because generative models are highly sensitive to prompt phrasing, a different prompt structure or a domain-specific clinical model could exhibit entirely different perturbation instability profiles, which was apparent when we used few-shot prompting and BioMistral-7B model in earlier experiments. Furthermore, the human-in-the-loop validation cohort is small ($N = 41$). While it provides a mathematically sound Cohen’s Kappa baseline for this study, a much larger annotated sample would be required to definitively validate the model’s accuracy on rare comor-

bidities. Additionally, the 0.95 SBERT cosine similarity threshold used to gate counterfactuals is a heuristic choice. While it successfully filters out catastrophic semantic failures, it may inadvertently drop subtle but clinically valid linguistic variations.

This project’s fairness scrutiny is also uneven across the pipeline. The mitigation methods in Section 4 get checked against an independent real-patient audit, bootstrap significance testing, and a second counterfactual construction at Level 2, precisely because we did not want to trust a single metric’s word for it. The cohort itself did not get the same treatment. Admissions missing any of the nine lab values are dropped rather than imputed (Section 3.1), on the reasoning that a filled-in value could itself encode bias. Section 6 reports that this exclusion is not random: Hispanic, Black, and Asian patients are missing at least one lab 33–35% of the time versus 24% for White patients, and non-Medicare patients are missing labs 34–42% of the time versus 16.5% for Medicare patients. What this project does not do is correct for that skew, every fairness number in Section 5 is computed on the surviving complete-case cohort as-is, with no reweighting or sensitivity analysis for how the exclusion itself might shift the reported metrics.

Similarly, the “Other/Unknown” race category, which shows the largest SHAP effect and the lowest baseline enrollment rate of any group (Section 3.4, Section 3.5), is a MIMIC-IV catch-all covering declined-to-state, genuinely unknown, and multi-racial patients rather than a coherent demographic group, and every fairness number this report reports for that category inherits that ambiguity. Neither gap changes the paper’s core findings about the mitigation methods, but a full accounting of this pipeline’s fairness would need to apply the same suspicion to how the cohort was built as this project applied to how it mitigated bias afterward.

The path-aware CDA pile split, real health versus unequal access, is a stated ethical assumption, not something derived from the data. Someone could reasonably split the thirty-five features differently, particularly prior admissions in the twelve months before the index visit, which we tested both ways precisely because we could not settle the question ourselves. The Level 2 VAE’s latent space is also not perfectly independent of the protected attributes, a probe recovers sex somewhat better than chance (Section 5.5), so Level 2 CFVR is reported with a noise-floor correction rather than treated as a clean ground truth; a tighter adversarial training objective on the VAE encoder, rather than the correction applied here after the fact, is a natural next step, particularly if Level 2 is ever extended to the unstructured text features from Stage 1, where a purely feature-split approach like path-aware CDA doesn’t obviously apply.

Every number in this report comes from a single seeded run of the pipeline, not from multiple runs averaged together. Fixing the seed, which this project did specifically to make the pipeline reproducible, guarantees that rerunning it produces the same numbers; it does not guarantee those numbers are representative of what a different seed would show. The point estimates in this report should be read as one realization of the pipeline, not as a robust estimate with an implicit error bar across runs. A proper multi-seed study, rerunning the full pipeline under five or ten seeds and reporting variance alongside each point estimate, is the natural next step and was outside this project’s scope.

Future work could also extend the DoWhy structural causal model beyond the two-node access pile used here to a fuller causal graph covering more of the thirty-five features, at the cost of needing to defend many more individual causal claims than the two this project relies on.

8 Conclusion

This project set out to test whether local fairness fixes in a clinical pipeline actually reduce end-to-end discrimination or just relocate it. The findings demonstrate that bias and instability enter the system long before the tabular classifier makes a prediction. During the Stage 1 text extraction, the LLM exhibited perturbation instability, altering extracted clinical risk factors purely in response to demographic token swaps. This upstream stochastic noise cascades directly

into the Stage 2 XGBoost model, actively shifting downstream probabilities and ultimately flipping Stage 3 care management enrollment decisions.

The pipeline-aware method, the version closest to that original question, produced a real but practically negligible effect on this run, and the effect is mildly harmful rather than helpful, a small increase in CFVR relative to baseline rather than a decrease, which is itself weak evidence for the bias migration hypothesis either way: a jointly-optimized fix did not obviously outperform fixing one stage at a time, in either direction.

The more decisive finding came from testing the standard baselines against each other and against four independent checks. Counterfactual data augmentation looked like the best method in the table by the metric it was evaluated on, a perfect zero, and none of those four independent checks, MPPD on real patients, the insurance axis, Level 2’s latent-space CFVR, or the race $\times$ sex and race $\times$ insurance intersections, back that claim up: on real patients its scores track baseline instead of the near-perfect consistency its own metric implies, on insurance it barely moves the needle, and at Level 2 and at the intersections it looks about the same as DoWhy SCM, the other method explicitly built to avoid the assumption CDA’s zero depends on.

Path-aware CDA and its DoWhy-based formalization both genuinely fix the individual-fairness problem CDA claims to fix on the axes where it matters most, with path-aware CDA additionally holding up on the insurance axis and the real-patient check where DoWhy SCM does not. Path-aware CDA is not a uniform winner, though: Level 2 and its intersectional replication are the one check where path-aware CDA is specifically the worst performer of the three, a genuine trade-off rather than a clean best choice. The practical recommendation coming out of this project is not “use this one method,” it’s that any fairness audit relying on a single metric, however principled that metric looks on paper, should be treated with the same suspicion this project ended up applying to CDA’s perfect zero. That recommendation should also be read alongside this project’s single-seed scope (Section 7): every comparison above is one realization of the pipeline, not a guarantee of what a different seed would show.

References

- [1] Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 2623–2631, 2019.
- [2] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- [3] Silvia Chiappa. Path-specific counterfactual fairness. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 7801–7808, 2019.
- [4] Moritz Hardt, Eric Price, and Nati Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NIPS)*, volume 29, pages 3323–3331, 2016.
- [5] Alistair E. W. Johnson, Lucas Bulgarelli, Lu Shen, et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10:1, 2023.
- [6] Faisal Kamiran and Toon Calders. Data preprocessing techniques for classification without discrimination. *Knowledge and Information Systems*, 33(1):1–33, 2012.
- [7] Niki Kilbertus, Mateo Rojas-Carulla, Giambattista Parascandolo, Moritz Hardt, Dominik Janzing, and Bernhard Schölkopf. Avoiding discrimination through causal reasoning. In *Advances in Neural Information Processing Systems*, volume 30, pages 656–666, 2017.

- [8] Matt J. Kusner, Joshua Loftus, Chris Russell, and Ricardo Silva. Counterfactual fairness. In *Advances in Neural Information Processing Systems*, volume 30, pages 4067–4077, 2017.
- [9] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [10] Razieh Nabi and David Benkeser. Fair risk minimization under causal path-specific effect constraints. *arXiv preprint arXiv:2408.01630*, 2024.
- [11] Razieh Nabi and Ilya Shpitser. Fair inference on outcomes. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.
- [12] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464):447–453, 2019.
- [13] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, 2019.
- [14] Darshali A. Vyas, Leo G. Eisenstein, and David S. Jones. Hidden in plain sight — reconsidering the use of race correction in clinical algorithms. *New England Journal of Medicine*, 383(9):874–882, 2020.
- [15] Tongshuang Wu, Marco Tulio Ribeiro, Jeffrey Heer, and Daniel S. Weld. Polyjuice: Generating counterfactuals for explaining, evaluating, and improving models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6707–6723. Association for Computational Linguistics, August 2021.
- [16] Javen Xu, Yeon-Mi Hwang, Samhita Kondareddy, Inés Dormoy, Serena Liang Jing, Malvika Pillai, Catherine Curtin, and Tina Hernandez-Boussard. Integrating group and individual fairness auditing in clinical ai: A post-hoc, model-agnostic approach. *medRxiv preprint*, 2025.
- [17] Brian Hu Zhang, Blake Lemoine, and Margaret Mitchell. Mitigating unwanted biases with adversarial learning. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, pages 335–340, 2018.

A LLM Prompt Template and Pydantic Schema

This appendix documents the Pydantic validation schema and the system prompt used for the Stage 1 Llama-3.1-8B-Instruct extraction pipeline.

A.1 Pydantic Validation Schema

The Python definition enforcing strict integer boundaries and token-level constraints via outlines:

Listing 1: Core Pydantic validation schema.

```
class RiskExtraction(BaseModel):
    comorbidity_burden_score: int = Field(ge=0, le=10)
    sdoh_evidence: str
    social_determinants_flags: List[str]
    psychiatric_complexity: str
    discharge_risk_indicators: List[str] = Field(max_length=3)
```

A.2 System Prompt Template

The exact prompt template passed to the model during batch inference:

Listing 2: System prompt template.

```
prompt_template = """<s>[INST] You are a strict clinical AI information
    extraction system. Read the clinical note and extract risk factors
    into a valid JSON object matching the schema.

SCHEMA FIELDS:
- comorbidity_burden_score: Integer (0-10), 1 point per chronic
    condition explicitly mentioned.
- sdoh_evidence: Short quote (max 15 words) proving the flags, or "no
    evidence" if none exist.
- social_determinants_flags: Array of strings chosen ONLY from:
    housing_instability, food_insecurity, substance_use,
    limited_social_support, unemployment, transportation_barrier. If
    sdoh_evidence is "no evidence", this MUST be [].
- psychiatric_complexity: Classify strictly based on the text as "low"
    (none/baseline), "medium" (managed anxiety/depression), or "high" (
    severe/active psychosis, bipolar, schizophrenia).
- discharge_risk_indicators: Array of max 3 short phrases extracted
    directly from the text representing primary diagnoses or
    complications. If none, output [].

RULES:
1. Base all extractions strictly on the clinical text provided. Do not
    assume or invent facts.
2. Medical conditions are NEVER social determinants.
3. If no social determinants are found, output "no evidence" and [].
4. Exact Mapping: The flags MUST match the evidence exactly. If
    evidence is "substance use", output ONLY ["substance_use"]. Do NOT
    infer housing instability, unemployment, or poor support from
    substance use.
5. Cognitive impairment (e.g., dementia, delirium, Alzheimer's, "poor
    historian") is a medical/neurological issue. Do NOT flag it as a
    social determinant or as psychiatric complexity.

CLINICAL NOTE:
{clinical_note_text}
```


Extract the JSON object now: [/INST]"

B Human Annotation Guidelines

This appendix documents the standardized evaluation criteria utilized for manual chart review and ground-truth annotation of the 41 validation discharge summaries. Annotators evaluated the raw clinical text blindly, without access to automated LLM predictions.

B.1 Annotation Rubrics

1. Comorbidity Burden Score: A global assessment of chronic physical disease severity. Annotators assigned an integer (0–10), allocating 1 point per distinct chronic condition explicitly documented in the medical history.

Table 9: Grading Rubric: Comorbidity Burden Score

Score	Severity	Clinical Criteria / Examples
0	None	No documented chronic systemic illnesses.
1–3	Mild	Single or well-controlled chronic illnesses (e.g., uncomplicated hypertension, diet-controlled diabetes).
4–6	Moderate	Multiple active chronic illnesses requiring ongoing therapy (e.g., insulin-dependent diabetes, moderate COPD).
7–10	Severe	End-stage or multi-system organ decompensation (e.g., NYHA Class IV heart failure, ESRD on hemodialysis).

2. Psychiatric Complexity: A categorical assessment grading the severity, chronicity, and acute stability of documented psychiatric comorbidities.

Table 10: Grading Rubric: Psychiatric Complexity

Level	Criteria
low	No psychiatric history, or well-controlled baseline conditions with minimal functional impairment.
medium	Active mood, anxiety, or trauma disorders requiring pharmacotherapy or active follow-up.
high	Severe active conditions, acute psychosis, severe bipolar disorder, schizophrenia, or acute psychiatric hold.

3. Social Determinants of Health (SDOH): Annotators identified social risk factors strictly from the following controlled vocabulary: `housing_instability`, `food_insecurity`, `substance_use`, `limited_social_support`, `unemployment`, and `transportation_barrier`. Any positive flag required a verbatim text snippet (maximum 15 words) recorded in the `human_sdoe_evidence` field.

4. Discharge Risk Indicators: A maximum of 3 short phrases capturing primary post-discharge complications, high-risk medication adjustments, or acute clinical instability.

B.2 Inter-Rater Reliability Formulation

Inter-rater agreement between human annotations and automated model extractions for ordinal variables was quantified using Cohen’s Quadratic Weighted Kappa (κ):

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (1)$$

where P_o represents observed proportional agreement and P_e represents expected agreement by chance, penalizing magnitude disparities across the multi-point severity scale.

C Counterfactual Perturbation Examples

To isolate the effect of demographic markers without corrupting the surrounding medical narrative, we utilized fixed-template substitutions. Table 11 illustrates this methodology, demonstrating how factual clinical sentences are projected onto counterfactual demographic axes prior to SBERT semantic validation.

Table 11: Illustrative examples of template-based demographic perturbation.

Perturbation Axis	Factual (Original Text)	Counterfactual (Perturbed Text)
Race	The patient is a 64-year-old White male presenting with acute exacerbation of COPD.	The patient is a 64-year-old Black male presenting with acute exacerbation of COPD.
Sex	She declines to provide a health care proxy, stating she has no family in the area.	He declines to provide a health care proxy, stating he has no family in the area.
Insurance	Patient is currently enrolled in Medicaid and reports difficulty affording insulin copays.	Patient is currently enrolled in Private Insurance and reports difficulty affording insulin copays.

D XGBoost Hyperparameter Search Space

To ensure the baseline Stage 2 tabular classifier was optimally tuned before fairness interventions were applied, hyperparameter optimization was conducted using Optuna. The search spanned 100 trials optimizing for validation Area Under the Receiver Operating Characteristic (AUROC). Table 12 details the search space and the final parameters selected for the main baseline model.

Table 12: Optuna hyperparameter search space and final selected values for the XGBoost classifier.

Hyperparameter	Search Space	Final Selected Value
n_estimators	[200, 600] (Integer)	452
learning_rate	[0.01, 0.1] (Uniform)	0.0208
max_depth	[4, 8] (Integer)	8
subsample	[0.6, 1.0] (Uniform)	0.6778
colsample_bytree	[0.6, 1.0] (Uniform)	0.9510
min_child_weight	[1, 10] (Integer)	10
scale_pos_weight	Fixed (Negative / Positive ratio)	3.710

E Complete Tabular Feature Dictionary

The Stage 2 readmission classifier utilizes 35 structured features derived directly from the MIMIC-IV relational tables (independent of the Stage 1 LLM extractions). Table 13 categorizes all 35 baseline features utilized in the modeling pipeline.

Table 13: Dictionary of the 35 MIMIC-IV tabular features used in the Stage 2 classifier.

Category	Included Features
Demographics (4)	Age (continuous), Sex (binary), Race (categorical encoded), Insurance Type (categorical).
Admission & Utilization (4)	Admission Type (categorical), Length of Stay (continuous days), Total ICU Length of Stay (continuous days), Number of ICU Stays (integer).
Historical (1)	Prior Admissions in Past 12 Months (integer).
Laboratory Values (9)	Bicarbonate, Blood Urea Nitrogen (BUN), Creatinine, Glucose, Hemoglobin, Platelets, Potassium, Sodium, White Blood Cell Count (all continuous, measured at admission).
Comorbidities (17)	Comorbidity Count (integer), plus 16 binary ICD-derived flags: Congestive Heart Failure, Arrhythmia, Hypertension, Chronic Pulmonary Disease, Diabetes, Renal Failure, Liver Disease, Cancer, Obesity, Depression, Anxiety, Alcohol Abuse, Drug Abuse, Psychosis, Coagulopathy, and HIV/AIDS.

F Baseline Cohort Characteristics

Table 14 outlines the baseline epidemiological and clinical characteristics of the MIMIC-IV patient cohort analyzed in this study, stratified by racial demographic groups prior to any fairness mitigation.

Table 14: Baseline cohort characteristics stratified by race. Continuous variables are presented as Mean (Standard Deviation) and categorical variables as Count (Percentage).

Characteristic	White	Black	Hispanic	Asian	Other/Unknown
Total Admissions (N)	41320	9102	3097	2008	4864
Demographics					
Age, years	64.4	59.9	55.0	61.6	60.0
Female Sex (%)	48.1	57.1	51.7	48.6	45.4
Clinical Severity					
Comorbidity Count	2.53	2.88	2.45	1.98	2.47
Prior Admissions (12m)	1.22	1.76	1.51	1.16	0.66
Length of Stay (days)	5.82	5.49	5.32	5.92	7.43
Number of ICU Stays	0.23	0.18	0.17	0.22	0.46
Outcomes					
30-Day Readmission Rate	21.4%	23.4%	22.9%	21.7%	13.3%
Stage 3 Enrollment Rate	9.8%	14.0%	12.3%	10.0%	5.2%